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ORIGINAL PAPER - FLUIDS, PLASMA AND PHENOMENOLOGY

Development of plasma sources and diagnostics for the simulation of fusion edge plasmas

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Abstract

Although the research on divertors and scrape-off layers (SOLs) has been not as focused on as the recent success of the Korean fusion program, a few linear plasma devices have been developed for simulating divertor and SOL plasmas: (1) diversified plasma simulator (DiPS), a versatile linear machine, has been developed for simulations of divertor and space plasmas with various electric probes, such as single, triple, and Mach Probes and gridded energy analyzer. DiPS consists of two major parts: a divertor plasma simulator with a LaB_6 DC plasma source and a space plasma simulator with a helicon RF plasma source, (2) divertor plasma simulator-1 (DiPS-1) is a part of DiPS with only a LaB_6 cathode, where a low-power laser-induced fluorescence (LIF) is added and more electric probe diagnostics are augmented; it is dedicated only for fusion edge and divertor plasmas, (3) Divertor Plasma Simulator-2 (DiPS-2) has been modified from the DiPS-1 by adding a magnetic nozzle with a limiter structure and by removing the helicon source and space chamber. DiPS-2 is a linear plasma device with a 4-inch LaB_6 cathode, the same as DiPS-1, and it is focused on the development of various diagnostics, such as those used for LIF and laser Thomson scattering (LTS) along with various electric probes, on the divertor and scrape-off plasmas and on the plasma-material interaction (PMI) research, such as that of tungsten and graphite as plasma-facing components (PFCs), (4) A Multi-Purpose Plasma (MP^2) device is a renovation of the Hanbit mirror device [Kwon et al., Nucl. Fusion **43**, 686 (2003)] with the installation of two plasma sources: LaB_6 (DC) and helicon (RF) plasma sources. A honeycomb-like large-area LaB_6 (HLA-LaB_6) cathode has been developed for the divertor plasma simulation to improve the resistance against the thermal shock fragility for large (8-inch) and high density plasma generation, (5) DiPS-2 has been augmented by adding another cylindrical device, called the Dust interaction with Surfaces Chamber (DiSC) for the generation and diagnostics of dusts. This combined system (DiPS-2+DiSC) has added two more diagnostics: Laser Photo-Detachment (LPD) for dust density and laser Mie Scattering (LMS) for dust size. Moreover, dusts or negative ions have been analyzed by using electric probes and capacitive diagram gauges in Transport and Removal of Dusts (TReD) device.

Keywords Plasma–material interaction · LaB_6 cathode · Divertor plasma · Plasma diagnostics · Electric probes · Laser-induced fluorescence (LIF) · Laser Thomson scattering (LTS)

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1 Introduction

Because the plasma-material interaction (PMI) of fusion devices, which is complex and coherent [1, 2], is one of the key issues for the success of ITER and DEMO, one needs to understand the edge plasma state, material damage, and their mutual interactions. Generally, the PMI should be analyzed by studying the plasma properties in the scrape-off layers (SOLs) and divertor (or limiter) regions [3–5], material erosion and deposition [6, 7], impurity transport [8], fuel retention [9], and dust production and detection [10–12]. For the PMI, edge plasma experiments in fusion reactors are not a unique solution because not only expensive to accumulate sufficient data but also difficult to conduct the experiments under various conditions. Thus, laboratory simulations of PMIs are necessary to quantify the long-term results of plasma (or beam) irradiations on the plasma facing components (PFCs), and to repeat laborious and multiple procedures.

The linear plasma device, generally called the linear divertor simulator, is an alternative PMI simulation tool for future fusion reactors, and it can provide relevant results of edge phenomena, such as SOL and divertor plasma transport, PMI, retentions of hydrogen isotopes, etc. Although the target exposure parameters in a linear plasma device cannot completely match the boundary plasma conditions in a tokamak, i.e., the geometry of the magnetic field configuration, the PMI-relevant parameters of material exposure in present linear devices would be very similar to the conditions expected in the ITER divertor. The differences between a linear device and a fusion reactor can be compensated as follows: (1) Particle density (or particle flux), which is low by approximately one order of magnitude, would be compensated for by accumulating exposure time; (2) Incident ion energy (or ion temperature) would be compensated by negative target biasing; (3) Material temperature would be covered by external heating and cooling of the target; (4) Plasma species (or impurities) would be compensated for by using secondary gas injection; (5) Transient events, such as edge localized modes (ELMs), would be simulated by using high power laser pulses and/or by a pulsed plasma with a pulsed operation of the plasma source (and/or an additional pulsed plasma gun) with steady-state plasma source. For laboratory research on PMIs, one needs to quantify the long-term results of plasma exposure of the PFCs and to repeat multiple tedious procedures.

For edge plasmas and PMI research, many linear plasma devices have been developed: (1) In the USA, PISCES-A [13] and -B [14] (UCSD), TPE(INL) [15], DIONISOS (MIT) [16], MPEX (ORNL) [17]; (2) In the European Union (EU), PSI-2 (FZJ, Germany) [18], pilot-PSI (Eindhoven University of Technology, The Netherlands) [19], Magnum-PSI (FOM-DIFFER, The Netherlands) [20, 21], YLPD (former UMIST, York University, The United Kingdom (UK) [22], GyM (IFP-CNR, Italy) [23], LENTA (Kurchatov Institute, Russia) [24]; (3) In Asia, Gamma-10/PDX [25] and MAP-II [26] (Tsukuba University, Japan), NAGDIS-II [27] NAGDIS-PG (upgraded from NAGDIS-I) [28] and PS-DIBA (Nagoya University, Japan) [29], TPD-Sheet IV (Tokai University, Japan) [30], STEP (Beihang University, China) [31], CRAFT (IPP-USTC, China) [32], CIMPLE-PSI (CPP-IPR, India) [33], and MAGPIE (Australian National University) [34]. In Korea, several PMI facilities have been developed, such as DiPS [35–37](for edge plasma and material test) and TReD (for dust particle transport) (Hanyang University) [38], MP^2 [39] (for edge plasma and material test), PLAMIS-II (for material test) [40] (KFE), MAXIMUS (KAIST) (for edge plasma transport) [41], and AF-MPD thruster (KAERI) (for material test) [42].

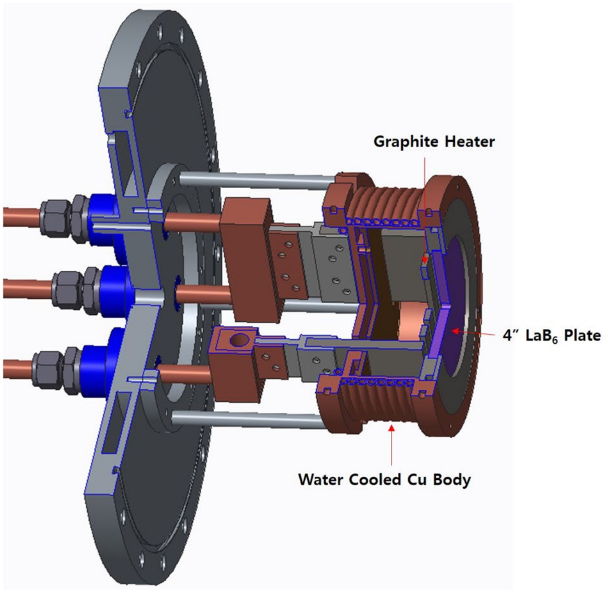
To perform the edge plasma and material test researches, two divertor plasma simulators have been developed: DiPS and MP^2. A rectangular large-area plasma device, called Transport and Removal of Dust (TReD), has been utilized for dust analyses. Several plasma diagnostics, such as electric probes, laser-induced fluorescence (LIF), laser Thomson scattering (LTS), laser photo-detachment (LPD) and optical emission spectroscopy (OES) for He collision-radiative model, have been developed to measure the properties of plasma and dust particles. In this paper, research on simulations of fusion edge plasmas is reviewed by using linear plasma devices in terms of devices and diagnostics, along with some experimental results.

2 Plasma devices for fusion edge plasma simulation

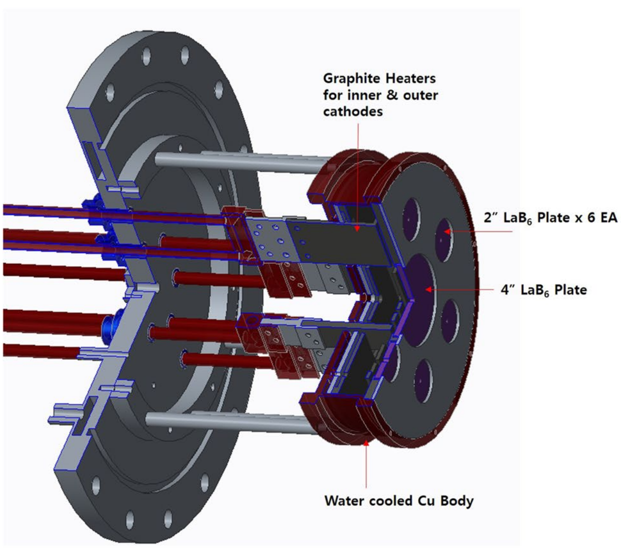
2.1 LaB_6 cathode for DiPS and MP^2 devices [36, 39]

Because the LaB_6 (lanthanum hexaboride) has a high thermal electron emission rate, low evaporation rate and high resistance to contamination in case of a vacuum break, it has been used to generate high density and high heat flux plasmas for linear divertor simulators and to produce high density electrons for high-power hall thrusters. Although LaB_6 is very weak against a thermal shock force, it can emit a higher numbers of electrons than other thermal electron sources, such as W (tungsten), BaO (barium oxide), etc.

Figure 1 shows the LaB_6 cathode developed for (a) DiPS -1 & -2 [36], and (b) MP^2 facility [39]. For DiPS-1 & -2, one uses a 4 inch diameter disk-type LaB_6 plate with a 1/4 inch thickness, which is indirectly heated by using a graphite heater. The graphite is more stable than tungsten and



(a)



(b)

Fig. 1 LaB_6 cathode to the (**a**) DiPS -1 & -2 and the (**b**) MP^2 linear divertor simulators

molybdenum for reaction problems with boron released from LaB_6. In addition, the graphite heater can be made larger (or thicker) than a tungsten heater due to its higher specific electrical resistivity, and the graphite heater is much stronger by mechanically and thermally than the tungsten heater. For heat transport in vacuum, mainly radiation, an important parameter is the view factor, which is defined as the ratio between the area of heat source facing the heated materials to the whole surface area of heater releasing the energy [43]. A graphite heater has larger view factor than tungsten, which is typically wire-type heater. For higher heating efficiency, one uses a tantalum foil of 0.2 mm thickness to shield the heat at back and the side of the heating region. The heating power to ignite the plasma is about 5 kW (280 A, 18 V), and it can be lowered to 4 kW while maintaining the same plasma discharge current due to the additional heat from ion bombardment.

For the MP^2 device, the Honeycomb-like Large-Area LaB_6 (HLA-LaB_6) cathode was developed for large area plasma generation [39]. The HLA-LaB_6 is composed of one inner LaB_6 cathode with a 4 inch diameter and six outer LaB_6 cathodes with 2 inch diameter. The large area LaB_6 cathode, which is composed of small pieces of LaB_6, can generate a large area plasma by reducing the cathode damage caused by thermal shock. The HLA-LaB_6 cathode array is separately heated by using inner and outer graphite heaters, and the spatial plasma profile can be partially controlled. Although Akhmetov et al. [44] recently reported that HLA-LaB_6 cathode is not capable of producing on high-density homogeneous plasma stream due to the large space between the LaB_6 plates, the HLA-LaB_6 of the MP^2 facility could control the plasma profiles because the cathode is located at a null point of the magnetic field and also the plasma resistivity in the source region could be changed by controlling the powers of the inner and the outer heaters. If one does not adopt a cusp magnetic field, the comment of Akhmetov et al. [44] would be plausible. In addition, one can change the geometry of the HLA-LaB_6 cathode to one with the same size and hexagonal shape of the LaB_6 plate if the MP^2 has not been shut down due to re-structuring the relevant organization. It is the reason that cathode is called a honeycomb-like cathode. The most important aspects of a HLA-LaB_6 cathode is that a large area plasma can be generated by combinations of small pieces of LaB_6 to reduce the thermal shock problem and to control the plasma profile.

2.2 Divertor plasma simulators (DiPS-1 and DiPS-2) [36, 37]

Diversified Plasma Simulator (DiPS) [36], a versatile linear plasma device, consists of three major parts: a divertor plasma simulator-1 (DiPS-1) with a LaB_6 DC plasma source, a space propulsion experiment and processing plasma simulator with a helicon RF plasma source, and a space plasma simulator with a low magnetic field. Figure 2 shows a schematic drawing of the DiPS. The space propulsion and processing plasma simulator consists of a quartz tube with an inner diameter of 32 mm (thickness of 2 mm) and m=+1 helicon antenna, an intermediate chamber with an inner diameter of 95 mm and a length of 375 mm, and a plasma acceleration chamber with an inner diameter of 185 mm and a length of 405 mm. The magnetic fields are about 500 G at the helicon antenna and a maximum of 900 G at the end of the quartz tube (intermediate chamber). The space simulator consists of a chamber with a diameter of 600 mm

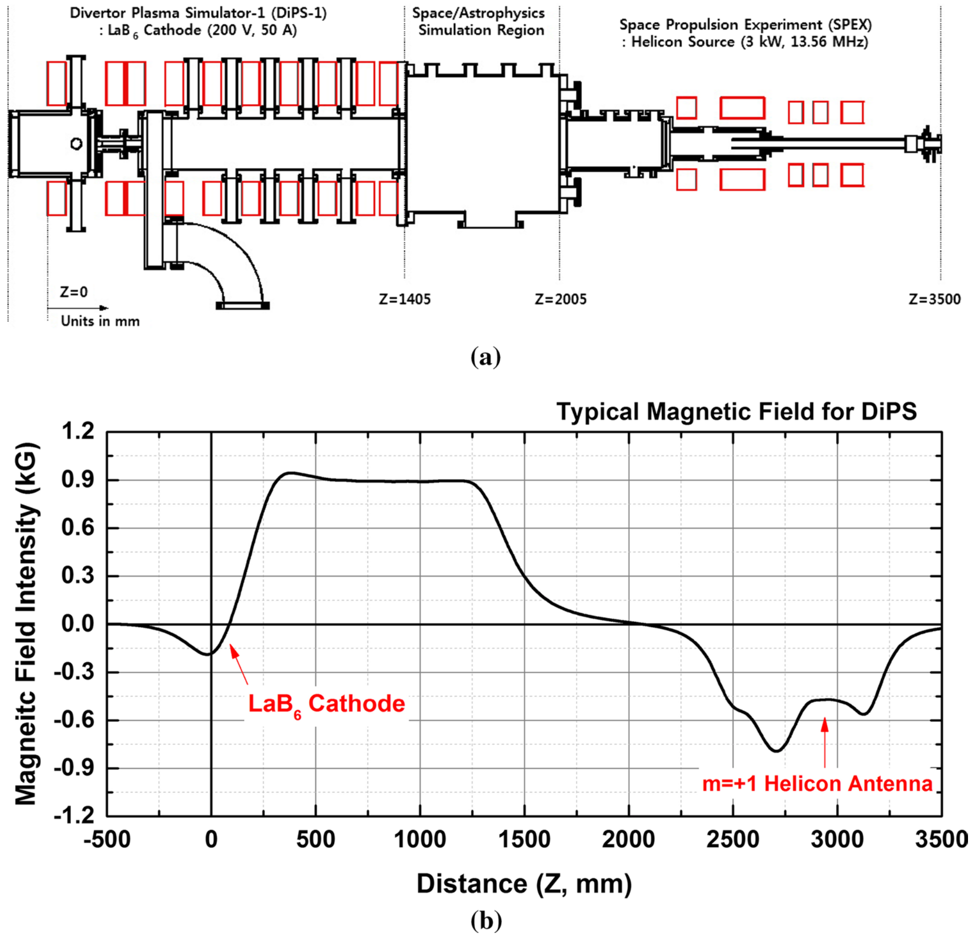


Fig. 2 DiPS and DiPS-1: **a** Schematic diagram of DiPS, which is composed of three parts: a divertor simulator with LaB_6 cathode (left side), which is called DiPS-1 later with augmented diagnostics for the divertor plasma simulations; a space simulator (middle part without a magnetic field); a space propulsion simulator with a helicon source (right side). **b** Typical magnetic field intensity along the *Z*-direction

and a length of 630 mm. A turbo pump with a pump rate of 1000 l/s is used to produce the vacuum. This simulator is installed on a rail system so that it can be easily moved to implement any experimental device or component in the test chamber of divertor and the processing simulators or the space simulator region. The space simulator can be used for artificial satellite experiments, as well as for space propulsion experiments. The plasma extracted from the divertor or processing simulator is confined in the space chamber after a reduction in the density by using grid electrodes.

DiPS-1 is part of the divertor plasma simulator of DiPS with only a LaB_6 plasma source, where a low-power LIF is added and more electric probe diagnostics are augmented. It is dedicated only to the fusion edge and divertor plasmas. The LaB_6 source chamber has a diameter of 260 mm and a length of 255 mm. Three magnets are located in front of the LaB_6 surface to generate cusp field (or null point or x-point). The cusp field is formed to confine of thermal electron on the LaB_6 disk (diameter of 101.6 mm) to the anode hole size (diameter of 28 mm). The cusp field disperses the heat load from the plasma to the entire surface of the LaB_6 and strengthens the electric field between the anode and cathode by reducing the distance between them. The source chamber has a double-wall structure for efficient water cooling, which is essential due to the high heating power of LaB_6 (~5 kW). The test chamber of the divertor simulator has a diameter of 200 mm and a length of 1007 mm. A turbo pump with a pump rate of 500 l/s is attached to maintain the pressure, and seven magnets are attached to produce a uniform magnetic field of up to 2 kG. Main goal of this test chamber is to simulate divertor edge plasmas. Floating and anode electrodes are inserted between the source and the test chamber. Due to the small hole (28 mm) in the floating and the anode electrodes, differential pumping of source region is maintained to generate plasmas.

DiPS-2 [37] is a modification of DiPS-1 to focus on divertor plasma physics and on plasma-material interaction research by changing the vacuum chamber and the magnetic field geometry, increasing the pump rates of the of vacuum pumps, and increasing the discharge powers. Figure 3 shows a schematic diagram of DiPS-2. DiPS-2 is clearly separated with five sections, the source region, extraction region, magnetic nozzle region, diagnostics region, and neutralization region. The feature that is most differed from DiPS-1 is that the neutral pressure can be controlled on the diagnostics and neutralization region without severely changing the

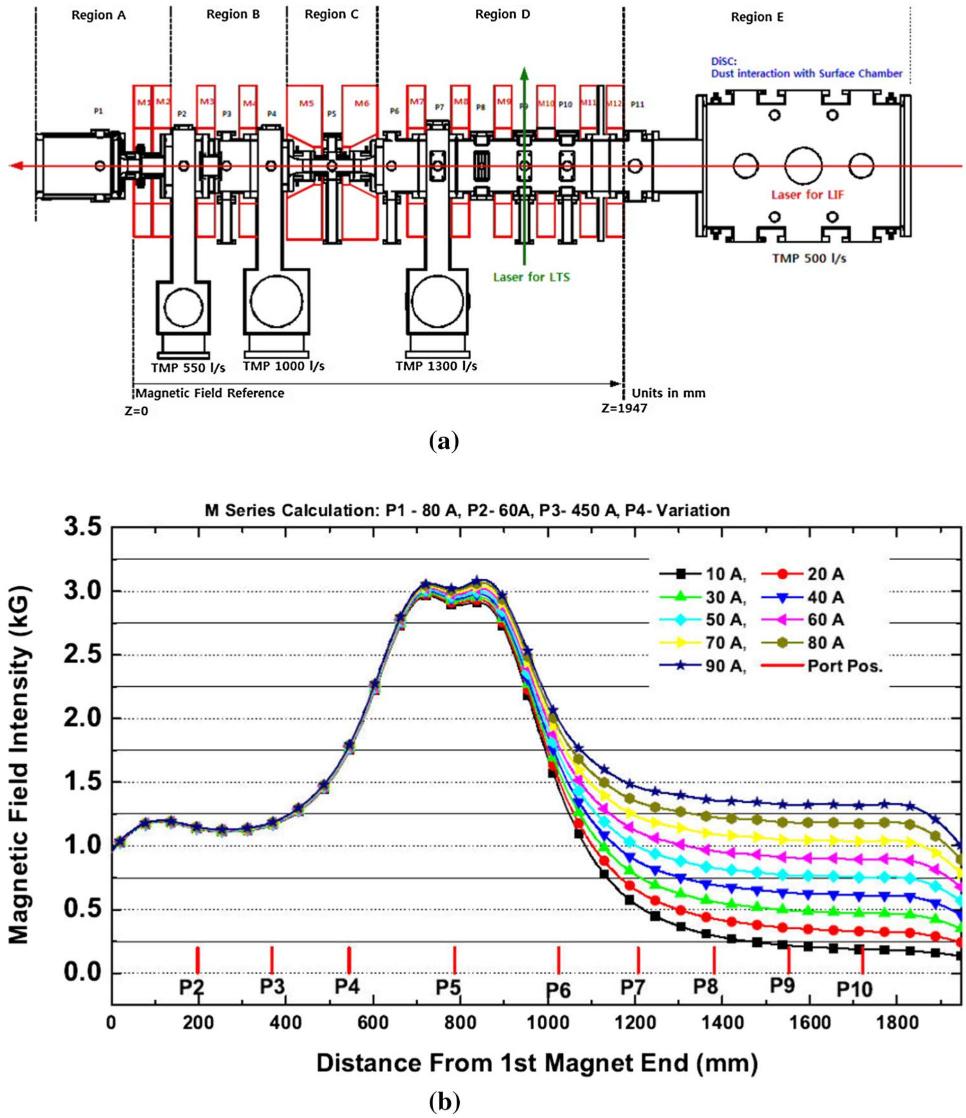


Fig. 3 DiPS-2: **a** Schematic diagram of the DiPS-2, where DiSC is augmented on the right side. **b** Typical magnetic field intensity along the *Z*-direction

pressures of other regions. One can increase the neutral pressure by secondary gas injection from 1 to over 100 mTorr in the diagnostics region without changing any other conditions while the pressures of the source and the extraction regions can be changed by less than 1 mTorr according to a baratron gauge. Because the plasma parameters before the diagnostics region remains the same, one can simulate the effect of an impurity (by using secondary gas injection) on the reduction in the heat flux. From experience in MAP-II and NAGDIS-II during the visit for collaboration, the neutral pressure can be controlled by closing the vacuum pump at end position, the plasma can be detached using secondary gas injection to the end position. However, the neutral network is changed with the pump down of the end position, and analyzing the diagnostics during the detached plasma is difficult due to changing references.

In addition, the DiPS-2 has a magnetic nozzle region for controlling the plasma extraction into the diagnostics region without changing the source conditions; the maximum magnetic field intensity is ~ 3.5 kG at the magnetic nozzle. DiPS-2 has been augmented by adding another cylindrical device, Dust interaction with Surfaces Chamber (DiSC) for the generation and the diagnostics of dusts. This combined system (DiPS-2+DiSC) has added two more diagnostics: Laser Photo Detachment (LPD) for dust density and laser Mie Scattering (LMS) for dust size. Figure 3b shows the typical magnetic field for DiPS-2, which is from the first electromagnet; the allowable ports can also be seen in the figure. DiSC is also added at the right side of the figure. The typical parameter of DiPS-1 and DiPS-2 are shown in Table 1.

2.3 MP^2: Multi-purpose plasma facility [39]

MP^2 facility [39] has been renovated from a Hanbit mirror device by adopting the same philosophy as that of DiPS-1

Table 1 Typical parameters for the DiPS-1, DiPS-2, scrape-off layer (SOL), and divertor region of fusion devices

Parameters	DiPS-1	DiPS-2	SOL and Divertor in Fusion
Magnetic field intensity (B)	1 kG	3.5 kG at nozzle	~ 5 T
		1.5 kG at diagnostics	–
Discharge voltage (V_{DIS})	up to 200 V	up to 200 V	–
Discharge current (I_{DIS})	up to 50 A	up to 150 A	–
Neutral pressure (P_n)	~ 200 mTorr at source	~ 50 mTorr at source	~ 0.1 mTorr at SOL region
	~ mTorr at diagnostics	0.5 – 100 mTorr at diagnostics	1- 30 mTorr at Divertor
Plasma density (n_p)	10^{13} cm^{-3} at diagnostics	~ 10^{14} cm^{-3} at diagnostics	10^{12} – 10^{14} cm^{-3}
Electron temperature (T_e)	~ 3 eV for Ar plasma	~ 3 eV for Ar plasma	1–100 eV
	~ 7 eV for He plasma	~ 7 eV for He plasma	
Ion temperature (T_i)	~ 0.1 eV for Ar	~ 0.1 eV for Ar	$T_i \approx T_e$
Particle flux (Γ)	< 10^{23} m^{-2} s^{-2}	10^{22} – 10^{24} m^{-2} s^{-2}	10^{22} – 10^{25} m^{-2} s^{-2}
Ion-neutral collision mean free path (λ_{mfp})	0.5–5 cm	0.05–10 cm	0.05–20 cm

[36] for a divertor plasma, space propulsion, and astrophysical research. This is done by installing two plasma sources, LaB_6 and helicon plasma sources, and by using three distinct simulators, a divertor plasma simulator, a space propulsion simulator, and an astrophysics simulator. However, it has been shut down due to a re-structuring of the relevant organization. Although the basic design concept of MP^2 is adopted from the DiPS-1, the MP^2 has unique features such as (i) a very large LaB_6 cathode, (ii) two magnetic field regions - high field (> 3 kG) and simple mirror regions, (iii) flexibility for future space/astrophysical experiments with a helicon source and with a large space and various diagnostics as a national plasma user facility in Korea. The MP^2 will carry on its own experiments related to future fusion divertor science, yet it may be open to the expected users depending upon their proposal with a provision of large spectrum of plasma parameters: (i) source type (dc and rf), (ii) plasma size (> 25 cm diameters), (iii) density (10^6–10^{13} cm^{-3}), (iv) electron temperature (0.1–10 eV), and (v) magnetic field (0–3 kG). Various diagnostics (electric probes, microwave interferometer, laser Thomson scattering, optical emission spectroscopy, etc) may also be provided. For MP^2, the divertor simulator with a HLA-LaB_6 cathode was developed during the 1st stage. Figure 4 shows a schematic diagram and magnetic intensity profile for MP^2, especially the divertor plasma simulator. The divertor plasma simulator is composed of the plasma source region, central cell region, and expansion region. The divertor plasma simulator of MP^2 has 19 electromagnets, which are numbered M1 to M19 starting from the LaB_6 cathode.

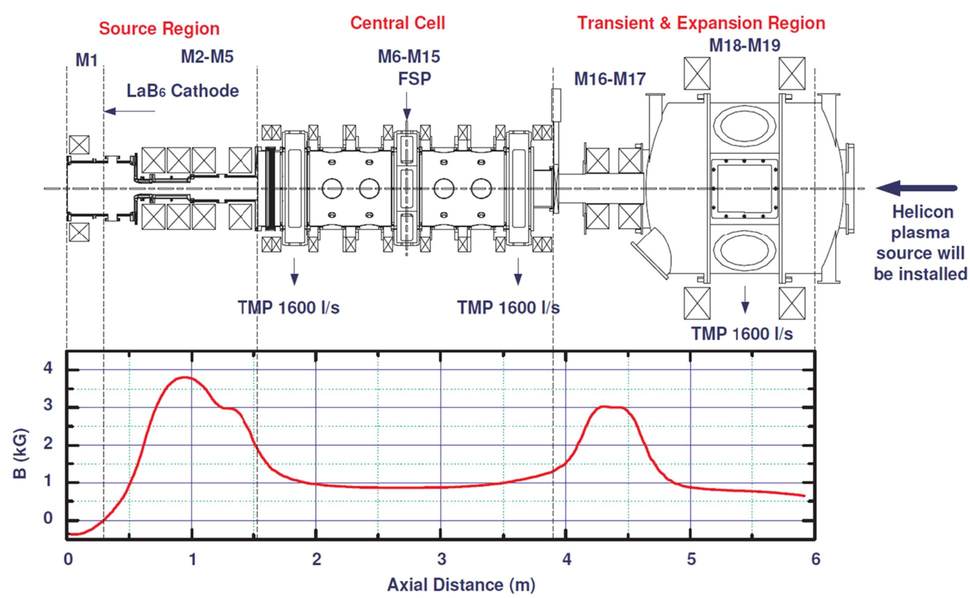


Fig. 4 Schematic diagram of MP^2 facility and its typical magnetic field intensity profile

2.4 TReD : Transport and removal experiments of dust [38]

Figure 5 shows a schematic diagram of TReD, which was developed to validate the dust particle controls in fusion plasmas. The TReD is composed of a large area inductively coupled plasma (ICP) source, a three-phase electrostatic curtain electrodes, and a dust collector. The background plasma for dust charging can be generated by using an ICP discharge with an RF power of less than 100 W (Max. 3 kW) at 13.56 MHz. For highly uniform plasmas, one designs the RF antenna (four linear lines) with a ladder shape to have the for same impedance on each line. The three-phase electrostatic curtain is made by using stainless steel for high mechanical strength without any insulator coatings in consideration of the acceptability in fusion devices. The dust transport length is about 1 m, which was used to be the minimum requirement for ITER level fusion devices. Dust or negative ions have also been analyzed by using electric probes and capacitive diagram gauges.

3 Plasma diagnostics for linear plasma device

3.1 Electric probes

The electric probes of the DiPS, such as single, triple, Mach and emissive probes, are generally operated on a fast scanning system for data collection without perturbating the plasma and damaging the probe itself with respect to the validity of the probe data. Figure 6 shows a fast scanning probe system in which probe tip can be driver at higher velocities up to 2 m/s. The fast scanning probe system consists of a pneumatic cylinder, a solenoid valve for activating the pneumatic cylinder, an optical switch for end position detection, vacuum bellows, and a linear position transducer for position monitoring. The probe assembled has a length of 900 mm, with a stroke of 300 mm. The pneumatic driver is operated at a pressure of 0.7 MPa, which is under the maximum allowable pressure of 0.8 MPa. Probe scan speeds of ~ 1.93 m/sec can be achieved.

In DiPS-1 & -2, the one of main research topics is the Mach probe's calibration with laser-induced fluorescence (LIF). Figure 7 shows the relationship between the plasma flow velocity measured by using the Mach probe and the LIF in the DiPS-1. If one uses the prevailing collisionless Mach probe models [45, 46], $K_M \sim 1.3$, the Mach number from Mach probe is overestimated by 10 to 30 % under the experimental conditions for comparing LIF measurements because the sheath density (or collected ion saturation current) may be affected by the momentum loss due to ion-neutral collisions and because at the same time the potential drop within the pre-sheath region is higher for accelerating ions up to the ion sound velocity. For considering the ion-neutral collisions, one uses momentum transfer collisions superimposed an ion-neutral charge exchange collision as $\sigma_{in} \sim 2 \times 10^{-18}\ \mathrm{m}^2$, which leads to $\tau \sim 0.017$ for ion-neutral collision and $\sigma \sim 10^{-4}$ for ionization of the non-dimensional parameters under the assumptions that the neutral temperature is $T_n = 1000$ K for $T_i = T_n$ and $T_n = 600$ K for $T_n \sim 0.5T_i$, which is based on the initial ion being taken from a Maxwellian distribution at room temperature $T = 300$ K. The resultant conversion factors for the Mach probe, K_M, are

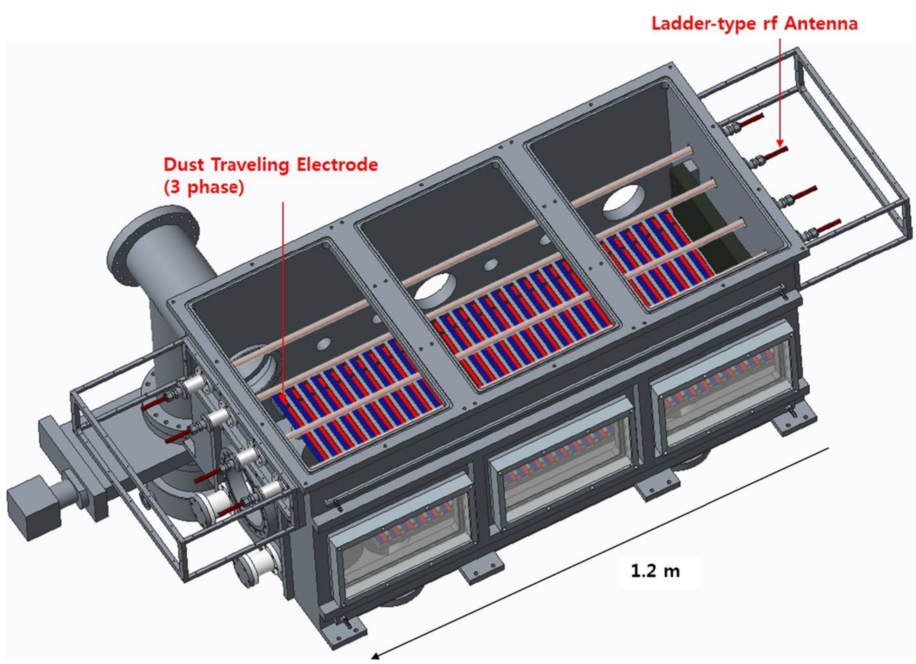
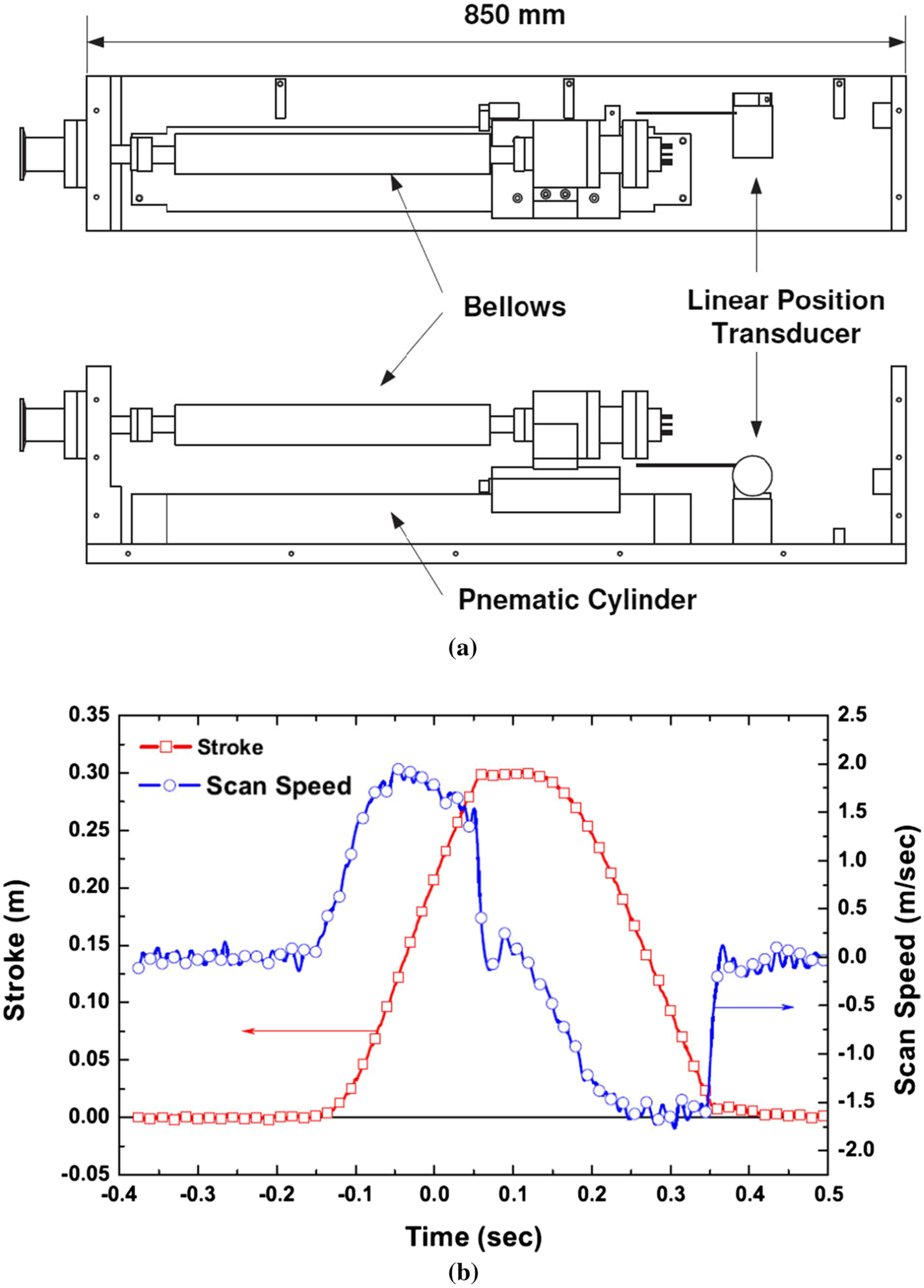


Fig. 5 Schematic diagram of the TReD device

Fig. 6 Fast probe driving system for the DiPS-1 and -2 **a** schematic diagram and **b** the scanning length and speed at the pneumatic driver pressure of 0.7 MPa



1.47 for $T_n = 1000$ K and 1.57 for $T_n = 600$ K, respectively. The results is more reliable than that for the collisionless case (see Fig. 7).

Because the DiPS-1 cannot generate the faster plasma flows, one apples the magnetic nozzle concept for the DiPS-2. In the DiPS-2, one can measure higher plasma flow velocity by using the Mach probe at a high density regime above 1×10^{13} cm^{-3}. However, the LIF could not occur in this regime due to the saturated population, i.e., due to rare meta-stable ions and the background noise, populating the meta-stable ions to excited levels is difficult. In the low density regime (less than $\sim 10^{13}$ cm^{-3}), the plasma flow velocity remains as low as 100 m/sec with both the Mach probe and the LIF. This velocity is related to a Mach number of 0.03, so one cannot perform the Mach probe calibration with LIF. This phenomena is acceptable because a Laval nozzle (like a magnetic nozzle) only works under isentropic condition (adiabatic and reversible), which means that the ion's energy loss is small or relatively low. That is, one can only generate a high velocity plasma flow under highly ionized condition, i.e.,

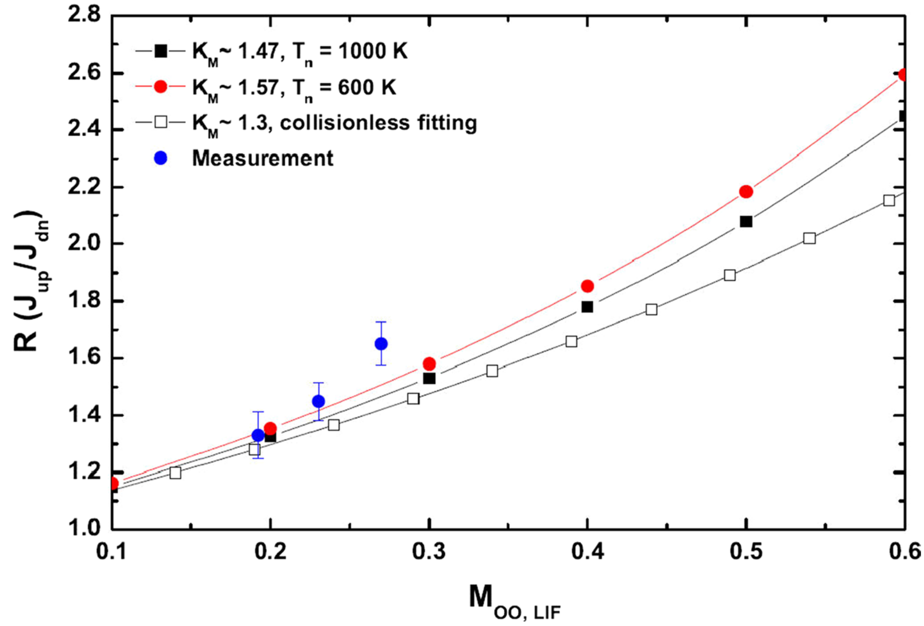


Fig. 7 Flow velocity measurements using a the Mach probe and a LIF. The ratio, R, is derived from the Mach probe and $M_{\infty,LIF}$ is from the LIF

the ion energy loss due to ion-neutral collisions is negligible. Figure 8 shows the Mach number are deduced by the Mach probe at position P5 (in Fig. 3). At the magnetic nozzle, a near sonic plasma flow is observed, although it is not confirmed by the LIF. Although the Mach probe calibration with LIF is not fully completed, one develops reliable probe theories with several atomic collisional effects, which can be applied under divertor plasma conditions. Although the ion drift resulting from scrape-off flow plays an important role in impurity transport and fluctuation levels, accurately measuring the flows is difficult due to various atomic processes, such as charge-exchange, recombination, and ionization.

3.1.1 Mach probe: effects of ionization and recombination in magnetized plasmas [47]

In this part, one introduces a fluid model for ion collection, including recombination and ionization, to deduce the Mach numbers in strong magnetic field. For Mach probe in magnetized, ionizing, and recombining plasmas, one starts with the Boltzmann transport equation for ions without volumetric sources, such as ionization and recombination, and assumes that $\mathbf{E} = E_z\mathbf{k}$ $\mathbf{B} = B\mathbf{k}$

$$v_z\frac{\partial f_i}{\partial z} + \frac{q}{m}E_z\frac{\partial f_i}{\partial v_z} = -S_t, \tag{1}$$

where the cross-field transport source can be approximated as [48]

$$S_t \equiv -v_x\frac{\partial f_i}{\partial x} \approx W\left[\alpha(f_{i\infty} - f_i) + (1-\alpha)(1-\frac{n}{n_\infty})f_{i\infty}\right], \tag{2}$$

with n_i, n_∞, f_i, $f_{i\infty}$, v_d (or v_z), W and α being the ion density along (or within) the pre-sheath, the ion density outside the pre-sheath, the atomic ion distribution along the pre-sheath, the atomic ion distribution outside the pre-sheath with drift velocity v_d, the flow velocity, the cross-field frequency ($\equiv D_\perp/a^2$, $D_\perp$ = anomalous cross-field diffusivity, a = probe radius) and the normalized shear viscosity ($\equiv \eta_\perp/n_i mD_\perp$), respectively. One can add ionization, recombination and charge exchange to Eqs. (1) and (2) for general cases.

For the case of a high electron temperature, generally $T_e > 2$ eV in hydrogen plasmas, one can consider the ionization source conditions are given by: (a) Bissel and Johnson (BJ) [49]: $S_i = <\sigma v>_{ion} n_e(z)f_{N\infty}(v)$ or (b) Emmert et al. (EE) [50]: $S_i = <\sigma v>_{ion} n_{eo}f_{N\infty}(v)|v|/C_s$, where $f_{N\infty}(v)$ is the distribution function for atomic neutrals. Although BJ's case seems to be reasonable, their model cannot recover the Maxwellian distribution in the limit $z \to \infty$ while EE's case can become Maxwellian although their ionization term does not seem to reasonable because no source particle with zero velocity. One choices BJ's case along with a cross-field transport source and obtains sheath values similar to those given by a kinetic model [48].

For recombination, there are two cases: one is electron-ion recombination (EIR)($S_r = -<\sigma v>_{rec} n_e f_i(v)$), which is general at low electron temperatures ($T_e < 2\,eV$) in hydrogen plasmas, and the other is molecular-activated recombination (MAR) ($S_r = -<\sigma v>_{rec} n_e f_M(v)$), which

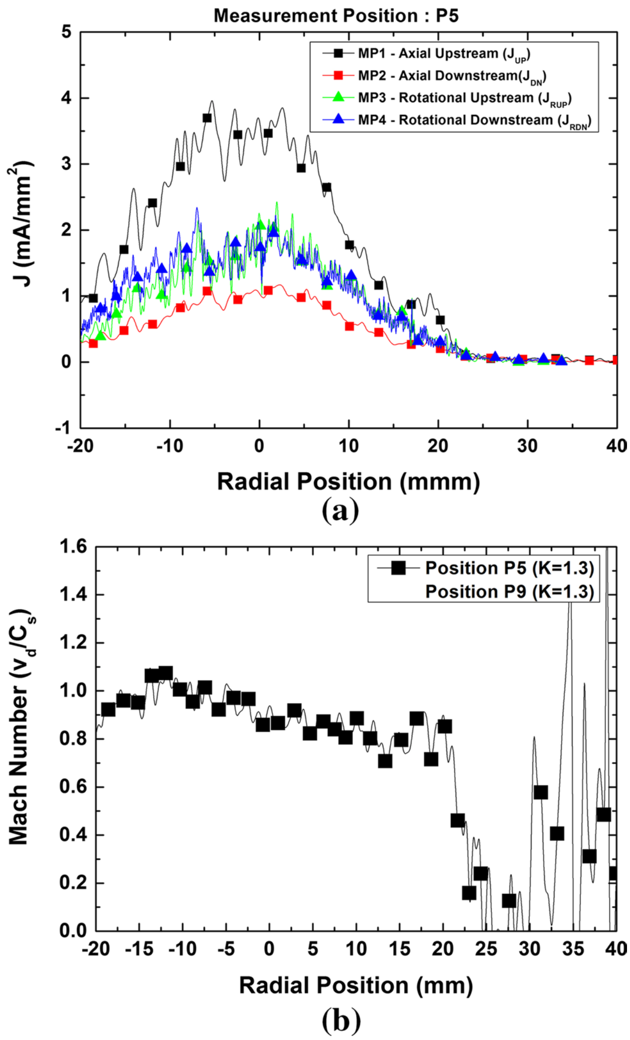


Fig. 8 Current density and Mach number deduced by using a Mach probe with an Ar plasma and a discharge current of 30 A at P5 in Fig. 3. **a** current density and **b** Mach number deduced by using calibration factor of $K = 1.3$

exists over a wide range of electron temperatures in hydrogen plasmas. Here, $f_M(v)$ is the distribution function for relevant molecular ions.

In a hydrogen plasmas, the dominant MAR processes are the followings: (i) dissociative attachment (DA: $H_2 + e \rightarrow (H_2^-)^* \rightarrow H^- + H$), followed by mutual neutralization (MN: $H^- + H^+ \rightarrow H + H^*$), (ii) ion conversion (IC: $H_2 + A^+ \rightarrow (AH)^+$), followed by dissociative recombination (DR: $(AH)^+ + e \rightarrow A + H^*$, and (iii) charge exchange ionization (CX: $H_2 + A^+ \rightarrow H_2^+ + A$), followed by dissociative recombination (DR : $H_2^+ + e \rightarrow H + H^*$). If one can ignore the negative hydrogen ions (neglecting DA), DR prevails. In that case, process (ii) is dominates than process (iii) [51]. Hence, the reaction rate of hydrogen MAR can be approximated as

$$< \sigma v >_{MAR} \approx < \sigma v >_{DR\leftarrow IC},$$

Because molecular ion density ($n_M \equiv n_i((\mathrm{AH})^+)$) is increased by IC while it is decreased by DR, one can further approximate the contribution of DR to MAR as

$$\begin{aligned}&< \sigma v >_{DR\leftarrow IC}\, n_e f_M \\ &\quad\approx (1-\delta) < \sigma v >_{DR} ((AH)^+ + e \rightarrow A + H^*) n_e f_M,\end{aligned}$$

where δ ($\equiv < \sigma v >_{non-IC} / < \sigma v >_{IC}$) is the ratio of non-IC process among IC, so that $1-\delta$ is the probability that DR occurs after IC. One considers the plasma conditions $n(H_2)/n_e \approx 0.1$; $T_e \approx T$ (heavy particles) $\approx 1\,eV$, ($T_e < 4 \sim 5eV$); $n_N \approx n_i$; and $< \sigma v >_{MAR} \approx 3 \times 10^{-16}$ m^3/s. For example, the MAR's rate is the dissociative recombination rates of the molecular ions for $10^{20} < n_e < 10^{21}$ m^{-3} [52, 53]. Then, one can deduce the source term by using atomic processes including MAR, EIR, and ionization, as

$$S_a = -K_M n_e f_M(v) - K_E n_e f_i(v) + K_I n_e f_N(v), \tag{3}$$

where f_M, f_i, and f_N are the distributions and reaction rates of molecular ions, atomic ions, and atomic neutrals, and K_M, K_E, and K_I are the reaction rates of molecular-activated recombination ($\equiv < \sigma v >_{MAR}$), electron-ion recombination ($\equiv < \sigma v >_{EIR}$, and ionization ($\equiv < \sigma v >_{ION}$), respectively.

For recombining and ionizing a plasma, the Boltzmann equation in Eq. (1) can be replaced by

$$\begin{aligned}v_z\frac{\partial f_i}{\partial z} + \frac{q}{m}E_z\frac{\partial f_i}{\partial v} &= S_t + S_a \\ &\approx W\left[\alpha(f_{i\infty} - f_i) + (1-\alpha)(1 - \frac{n}{n_\infty})f_{i\infty}\right] \\ &\quad + n_e\left[-K_M f_M - K_E f_i + K_I f_N\right].\end{aligned} \tag{4}$$

If one assumes that the temperature of atomic neutrals is the same as that of molecular ions, i.e., $T_N = T_M \equiv \tau T_i$, and $m_N = m_i$, the unperturbed distribution functions of ions ($f_{i\infty}$) and atomic neutrals ($f_N \equiv f_{N\infty}$) can be expressed as

$$f_{i\infty}(v) = n_\infty\sqrt{\frac{m_i}{2\pi T_i}}exp\left[-\frac{m_i(v-V_d)^2}{2T_i}\right],$$

$$f_N(v) = v_1 n_\infty\sqrt{\frac{m_i}{2\pi\tau T_i}}exp\left[-\frac{m_i v^2}{2\tau T_i}\right],$$

where $v_1 \equiv n_N/n_\infty$. By taking moments of Eq. (4) and using dimensionless parameters such as $L \equiv C_s/W(z)$, $y \equiv \int[W(z)/C_s]dz$, or $y \equiv zW/C_s$, $M \equiv V_z/C_s$, $n \equiv n_i/n_\infty$, one can arrive at the following dimensionless equations:

$$M\frac{\mathrm{d}n}{\mathrm{d}y} + n\frac{\mathrm{d}M}{\mathrm{d}y} = 1 - n + k_i n - k_r n^2, \tag{5}$$

$$\frac{\mathrm{d}n}{\mathrm{d}y} + nM\frac{\mathrm{d}M}{\mathrm{d}y} = (M_\infty - M)[1-(1-\alpha)n] - k_i nM, \tag{6}$$

where k_i, k_m, k_e and k_r are the normalized ratios of ionization [$\equiv (K_I Z n_N a/C_s)(L/a)$], molecular-activated recombination [$\equiv (K_M Z n_M a/C_s)(L/a)$], electron-ion recombination ($\equiv (K_E Z n_\infty a/C_s)(L/a)$) and total recombination ($\equiv k_m + k_e$) with respect to the cross-field transport contribution, respectively, and n_M is the density of relevant molecular ions. Here C_s is the ion acoustic speed [$\equiv \sqrt{(T_e + ZT_i)/m_i}$], n_N is the atomic neutral density, and quasi-neutrality ($en_e = Zen_i$) is used. After some arrangements, one can obtain the following equations for dn/dy and dM/dy:

$$\frac{\mathrm{d}n}{\mathrm{d}y} = \frac{M_\infty - 2M - (M_\infty - M)(1-\alpha)n + (1+k_r n)nM}{1-M^2}, \tag{7}$$

$$\frac{\mathrm{d}M}{\mathrm{d}y} = \frac{1 - n - M(M_\infty - M)[1-(1-\alpha)n] + k_i n(1-M^2) - k_r n^2}{n(1-M^2)}. \tag{8}$$

Dividing Eq. (7) by Eq. (8) leads to

$$\frac{1}{n}\frac{\mathrm{d}n}{\mathrm{d}M} = \frac{M_\infty - 2M - (M_\infty - M)(1-\alpha)n + (1+k_r n)nM}{1 - n - M(M_\infty - M)[1-(1-\alpha)n] + k_i n(1-M^2) - k_r n^2}. \tag{9}$$

3.1.2 Mach probe: effect of ion-neutral collision in un-magnetized plasma [54, 55]

In this part, one considers the effect of ion-neutral collision, especially ion-neutral momentum transfer collisions, on the Mach probe in un-magnetized plasmas. For this, one takes the moments of the kinetic equation and converts them to the fluid equations:

$$n\frac{\mathrm{d}v}{\mathrm{d}z} + v\frac{\mathrm{d}n}{\mathrm{d}z} = v_{iz}n + \frac{v_{ti}}{a}(n_\infty - n), \tag{10}$$

$$\begin{aligned} mnv\frac{\mathrm{d}v}{\mathrm{d}z} &= enE - k_B T_i \frac{dn}{dz} - mn(v_{iz} + v_{in})(v - v_d) \\ &+ \frac{v_{ti}}{a_p} m(n_\infty - n)(v_d - v), \end{aligned} \tag{11}$$

where v, E, T_i, k_B, v_{iz}, and v_{in} are the ion fluid velocity, the electric field along the pre-sheath, the ion temperature, the Boltzmann constant, the ionization frequency, and the ion-neutral collision frequency, with ion-neutral charge exchange and momentum exchange being given by $v_{in} = v_{cx} + v_m$, respectively, the electron density is assumed to follow the Boltzmann relation

$$n = n_\infty \exp\left(\frac{e\phi}{k_B T_e}\right). \tag{12}$$

When the dimensionless variables

$$\begin{aligned} &N \equiv n/n_\infty,\ \ x \equiv z/a,\ \ M \equiv v/C_s,\ \ M_\infty \equiv v_d/C_s, \\ &C_s \equiv \sqrt{(k_B T_e + k_B T_i)/m_i}, \\ &\tau \equiv v_{ti}/C_s, \\ &\rho \equiv v_{in}a_p/C_s, \\ &\sigma \equiv v_{iz}a_p/C_s,, \end{aligned} \tag{13}$$

are used, the governing equations become

$$N\frac{\mathrm{d}M}{\mathrm{d}x} + M\frac{\mathrm{d}N}{\mathrm{d}x} = \sigma N + \tau(1-N), \tag{14}$$

$$NM\frac{\mathrm{d}M}{\mathrm{d}x} + \frac{\mathrm{d}N}{\mathrm{d}x} = -N(\sigma+\rho)(M - M_\infty) + \tau(1-N)(M_\infty - M). \tag{15}$$

The left-hand side of the normalized momentum equation (Eq. 11) can be obtained from a balance of the ion inertial force ($mnv\mathrm{d}v/\mathrm{d}z$), the electrostatic force ($enE = -en(\mathrm{d}\phi/\mathrm{d}z)$) and the ion pressure gradient force ($-k_B T_i \mathrm{d}n/\mathrm{d}z$):

$$\begin{aligned} &mnv\frac{\mathrm{d}v}{\mathrm{d}z} - enE + k_B T_i\frac{\mathrm{d}n}{\mathrm{d}z} \\ &= \frac{mn_\infty C_s^2}{a_p}NM\frac{\mathrm{d}M}{\mathrm{d}x} + \frac{(k_B T_e + k_B T_i)n_\infty}{a_p}\frac{\mathrm{d}N}{\mathrm{d}x} \\ &= \frac{mn_\infty C_s^2}{a}\left(NM\frac{\mathrm{d}M}{\mathrm{d}x} + \frac{\mathrm{d}N}{\mathrm{d}x}\right), \end{aligned}$$

where $en(\mathrm{d}\phi/\mathrm{d}z) = k_B T_e(\mathrm{d}n/\mathrm{d}z)$ from the Boltzmann relation is used. These equations can be written as

$$\begin{aligned} \frac{\mathrm{d}N}{\mathrm{d}x} &= \frac{M_\infty - 2M}{1-M^2}\tau^* \\ &+ \frac{\sigma N(M_\infty - 2M) + \rho N(M_\infty - M)}{1-M^2}, \end{aligned} \tag{16}$$

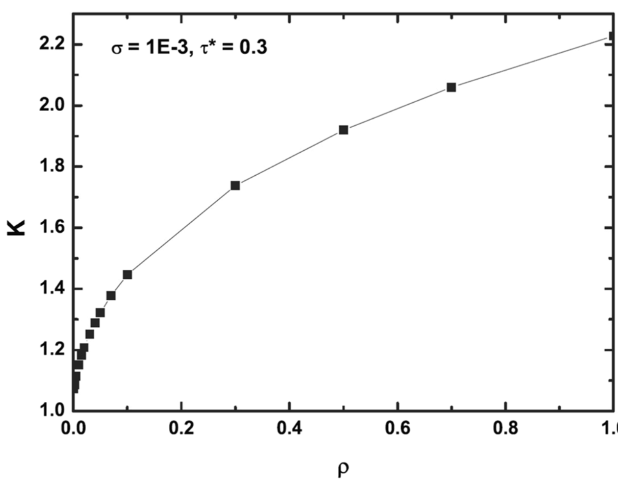
$$\begin{aligned} \frac{\mathrm{d}M}{\mathrm{d}x} &= \frac{(M^*+1)\tau^*}{(1-M^2)N} \\ &+ \frac{\sigma(M^*+1) + \rho M^*}{1-M^2}, \end{aligned} \tag{17}$$

where $\tau^* \equiv (1-N)\tau$ and $M^* \equiv M^2 - M_0 M$. If the collision terms in Eqs. 16 and 17 are not neglected, the solutions for $N(x)$ and $M(x)$ cannot be given by using an analytical method. However, $N(M)$ can be given in Stangeby's form [56] as

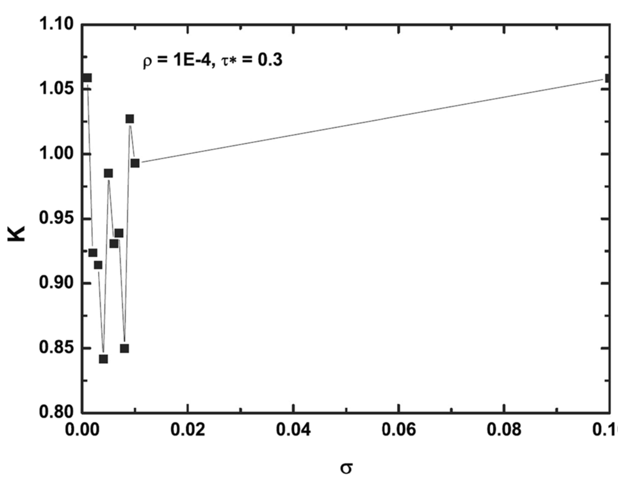
$$\frac{\mathrm{d}N}{\mathrm{d}M} = \frac{(M_0 - 2M)N\tau^* + \sigma N^2(M_0 - 2M) + \rho N^2(M_0 - M)}{(M^*+1)\tau^* + \sigma N(M^*+1) + \rho NM^*}. \tag{18}$$

If one numerically calculates Eq. (18) for specific conditions the normalized density (N), where no over shooting of N

($N > 1$) occurs contributions of ionization and ion-neutral collisions are small. The numerical result for Eq. (18) is shown in Fig. 9, which is the relation between the calibration factor (K) and the collision parameters such as ρ (normalized momentum collision frequency) and σ (normalized ionization collision frequency). When ρ or σ is fixed and when the other value is allowed to vary as in Fig. 10, the variation of K with ρ is bigger than the case for σ, which indicates that the collisional effect of ρ is more dominant than that of σ. Figure 10 shows the variation of K in terms of neutral pressures, neutral temperatures and ion temperatures. At higher neutral pressures of more than 15 mTorr, ρ is bigger than 0.03, and the effect of ρ is dominant, as in the case of Fig. 9.

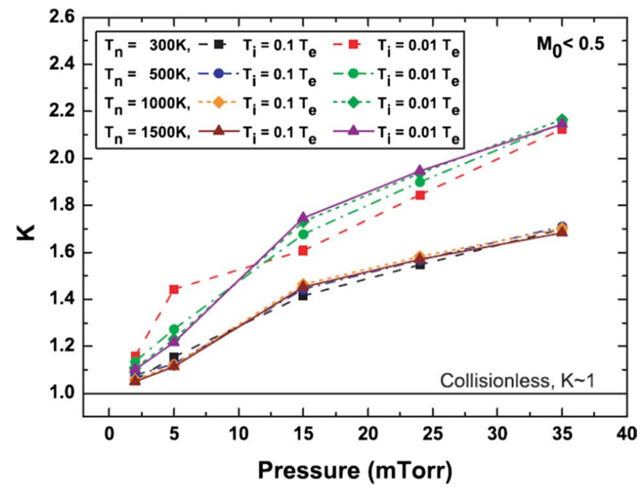


(a)

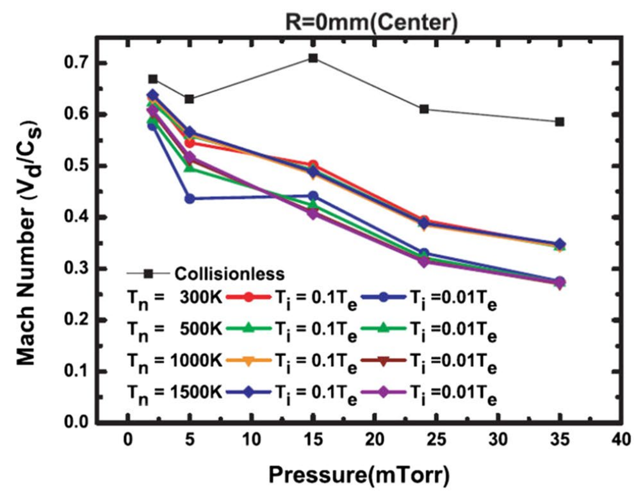


(b)

Fig. 9 Effect of the collision parameters on the calibration factor (K): for **a** momentum transfer, and for **b** ionization collision



(a)



(b)

Fig. 10 Effect of neutral pressures on the calibration factor: Variation in **a** K, and **b** Mach numbers deduced using collisionless and collisional theory from NAGDIS-experiment with the assumptions of neutral pressures, neutral temperatures and ion temperatures in a He plasma

3.1.3 Probe theory for electro-negative plasmas [57, 58]

When a strong negative bias voltage is applied to the probe such that $V_B \le -\max(|V_p|, |V_f|)$ [59] with arbitrary plasma potential, V_p, and floating potential, V_f, relative to ground, negative ions neither affect sheath's formation nor contribute to the (positive) ion saturation current. Thus, the positive ion saturation currents at pressures P_1 and P_2 are given by

$$I_{S+}(X_{1,2}) \propto A_{1,2}[N_+(X_{1,2})n_S(X_{1,2})][v_S(X_{1,2})\sqrt{T_e(X_{1,2})/M(X_{1,2})}], \tag{19}$$

where $A_{1,2}$ are sheath areas for the collection of ions, N_+ is the positive ion density, T_e is the electron temperature,

$X_{1,2}$ are gas mixtures of background(noble gas) and added negative ion gas (depending upon flow rate) at flow rates of $F_{1,2}$ (or pressure $P_{1,2}$), $M(X_{1,2})$ are the positive ion reduced masses at $F_{1,2}(P_{1,2})$. n_S and v_S are the normalized sheath density and sheath velocity, which are given as the following for Boltzmann negative ions and electrons:

$$n_S = [1-\alpha]\exp[-\eta_S] + \alpha(X)\exp[-\eta_s/t_m],$$
$$v_S = \sqrt{2[t_p+\eta_s]},$$

where α is the ratio of negative ions to positive ions ($\equiv N_-/N_+$, η_S is the sheath (Bohm) potential normalized to the electron temperature ($\equiv -eV_S(X)/T_e$), and t_+ and t_- are the temperatures of positive and negative ions normalized to the electron temperature, respectively.

From quasi-neutrality, the following should be true:

$$N_+(X_{1,2}) = N_e(X_{1,2}) + N_m(X_{1,2}), \tag{20}$$

where N_e and N_m are the densities of electrons and negative ions, respectively. The saturated collection currents (I_{SN}) of negative charges, electrons and negative ions can be approximated as the electron saturation current based upon the mass ratio and the temperature. Although one cannot know the exact form of the electron saturation current within a magnetic field, the saturation current for negative charges can be expressed by

$$I_{SN} \approx I_{SE}(X_{1,2}) \propto N_e(X_{1,2})A_p\sqrt{T_e(X_{1,2})/m_e}, \tag{21}$$

where A_p is the probe area for the collection of Boltzmann electrons. From Eq. (19) and (21), one simply finds that saturation current varies as

$$\frac{i_2}{\Omega_2} = \frac{N_2}{N_1}\sqrt{\frac{\tau_2}{\mu_2}},\ \epsilon_2 = \frac{N_{e2}}{N_{e1}}\sqrt{\tau_2}, \tag{22}$$

where $N_{1,2}$ and $N_{e1,e2}$ are the positive and the electron densities. The non-dimensional parameters of Eq. (22) are defined as

$$i_2 \equiv \frac{I_{S+}(X_2)}{I_{S+}(X_1)},\ \epsilon_2 \equiv \frac{I_{SE}(X_2)}{I_{SE}(X_1)},\ \tau_2 \equiv \frac{T_e(X_2)}{T_e(X_1)},$$
$$\mu_2 \equiv \frac{M(X_2)}{M(X_1)},\ \Omega_2 \equiv \frac{A_S(X_2)n_S(X_2)v_S(X_2)}{A_S(X_1)n_S(X_1)v_S(X_1)}.$$

Equation (20) then becomes

$$i_2 = \frac{\sqrt{\mu_2}}{\Omega_2} = \epsilon_2(1-\alpha_1) + \alpha_2\sqrt{\tau_2}, \tag{23}$$

and one obtains ratio of negative ion to positive ions as

$$\alpha_1 = 1 - \frac{i_2\sqrt{\mu_2}}{\epsilon_2\Omega_2} - \alpha_2\frac{\tau_2}{\epsilon_2}, \tag{24}$$

where $\alpha_1 \equiv N_{-1}/N_1$ and $\alpha_2 \equiv N_{-2}/N_2$, which mean that the ratio of negative ion to positive ion at P_1 and P_2, respectively. Hence i_2, ϵ_2, and τ_2 can be measured from I-V curves, but μ_2 (ratio of reduced masses), Ω_2 (ratio of sheath factors), and α_2 need to provided in order to get α_1.

If one considers cylindrical and planar Langmuir probes, the saturation currents of ions and electrons are affected by the sheath factor (due to a change in the collection area) and the geometrical factor, respectively [60, 61]. If one also considers a very thin cylindrical probe ($a_p/\lambda_D < 1 \rightarrow a_p/\chi_S \ll 1$), one can obtain two different sets of equations, are for α_1 and the other for α_2, where λ_D and χ_S are the Debye length and sheath thickness, respectively. Then, one can solve the equations for α_1 and α_2. If one treats that the Eq. (24) is deduced (or measured) by using a planar Langmuir probe, the ratio of negative ion to positive ion, deduced by using a cylindrical Langmuir probe, can be expressed as

$$\alpha_1 = 1 - \frac{i_{2C}\sqrt{\mu_2}}{\epsilon_{2C}\Omega_{2C}} - \alpha_2\frac{\tau_2}{\epsilon_{2C}}. \tag{25}$$

Here, ratios of the electron temperatures (τ) and the effective masses (μ) should remain the same for the same plasma even when different probes are used, but the ratios of the saturation currents (i_{2C}, ϵ_{2C}) and the sheath factors (Ω_{2C}) for cylindrical probes are different from those for a planar probe ($i_{2C}, \epsilon_{2C}, \Omega_{2C}$). Especially, the ratio, ϵ_{2C}, of electron saturation currents for a thin cylindrical probe is different from that of a planar probe, ϵ_2, due to the limited current caused by orbital motions when the sheath is thick compared to the probe radius [60]. Then, the $\alpha_{1,\,2}$ are calculated as

$$\alpha_1 = 1 - \frac{\sqrt{\mu_2}}{\epsilon_2 - \epsilon_{2C}}\left(\frac{i_2}{\Omega_2} - \frac{i_{2C}}{\Omega_{2C}}\right), \tag{26}$$

$$\alpha_2 = \frac{\sqrt{\mu_2/\tau_2}}{1-\epsilon_{2C}/\epsilon_2}\left(\frac{i_2}{\Omega_2} - \frac{i_{2C}}{\Omega_{2C}}\right) - \frac{i_2}{\Omega_2}\sqrt{\frac{\mu_2}{\tau_2}}. \tag{27}$$

Here, $i_{2,\,2C}$ and $\epsilon_{2,\,2C}$ can be obtained from two I -V curves, which are taken by using planar and cylindrical probes at two different pressures. However, one still needs the ratio of effective masses, $\mu_{2,\,2C}$, and the ratio of sheath factors, $\Omega_{2,\,2C}$, to calculate $\alpha_{1,\,2}$.

First, the effective mass of positive ions can be expressed as $M_1 = (n_{a1}M_a + n_{b1}M_b)/(n_{a1}+n_{b1})$, where $n_{a1,\,b1}$ and $M_{a,\,b}$ are the positive ion densities generated by the added gas for negative ion production and generated by the background gas, and the masses of the added and the background gases, respectively.

Then, the following are obtained with $\delta \equiv n_{a1}/N_{-1}$: $\mu_2 = [M_b + \delta\alpha_2(M_a - M_b)]/[M_b + \delta\alpha_1(M_a - M_b)]$, where $n_{a1}/N_1 = (n_a 1/N_{-1})(N_{-1}/N_1) = \delta\alpha_1$, and $n_{a2}/N_2 = \delta\alpha_2$, $N_{1,\,2} = n_{a1,2} + n_{b1,2}$, $n_{a1,\,2} = \delta n_{-1,\,2}$, $0 < \delta < 1$.

Second, if one assumes Boltzmann negative ions as in previous works [59, 62–65], the ratio of sheath factors, Ω_2, becomes

$$\Omega_2 = \frac{A_2\sqrt{\gamma t_{+2} + 2\eta_{S2}}[(1-\alpha_2)e^{-\eta_{S2}} + \alpha_2 e^{-\eta_{S2}/t_{-2}}]}{A_1\sqrt{\gamma t_{+1} + 2\eta_{S1}}[(1-\alpha_1)e^{-\eta_{S1}} + \alpha_1 e^{-\eta_{S1}/t_{-1}}]}. \tag{28}$$

The negative ion density measurement is performed in a SF_6 plasma, which is an in the inductively coupled plasma source with an rf power of 100 − 500 W (13.56 MHz). [58] The gas pressure is varied in the range from 0.4 to 1.0 mTorr. Figure 11 shows the negative ion density, electronegativity, and ratio of negative ions to electrons, which are deduced by using single Langmuir probe measurements. Physically, one of the three values of α contributes to the overall variation in the negative ion density with pressure. The electronegativity ($\alpha \equiv N_m/N_+$) and the ratio of the negative ion to the electron density ($\beta \equiv N_m/N_e$) which increase slightly with pressure. This behavior of β is qualitatively consistent with the experimental results for other electronegative discharges measured by Tuszewski [66] and Gudmundsson [67].

Generally, negative ions are produced through molecular dissociation and electron attachment (electron affinity energy): the former process requires a relatively high electron energy, ~ 5 eV for O_2 plasma, while the latter is only possible for an energy of 1.4 eV for O^-, O_2^-, O_3^-, and other ions. However, the details of negative ion generation and its effect on the plasma (or electron) are not yet fully understood. For this, probe diagnostics have been performed in Ar and O_2 mixture plasmas, generated by a filament discharge and with an electron temperature under 5 eV. The electron temperature can be deduced by using the both general I-V characteristics and the electron energy distribution function (EEDF) (and/or electron velocity distribution function (EVDF)). Although the electron temperature is less than that for molecular dissociation, from the analyses, one can confirm that negative ions can be generated by both molecular dissociation due to high energy electrons (tail part) and electron attachment due to low energy electrons. Figure 12 shows the results of EEDF and EVDF. With oxygen flow, the full width half maximum (FWHM) of EVDF is narrower and the peak velocity is also slower. Figure 13 shows that the ratio of the electron density to the ion density and the electro-negatively.

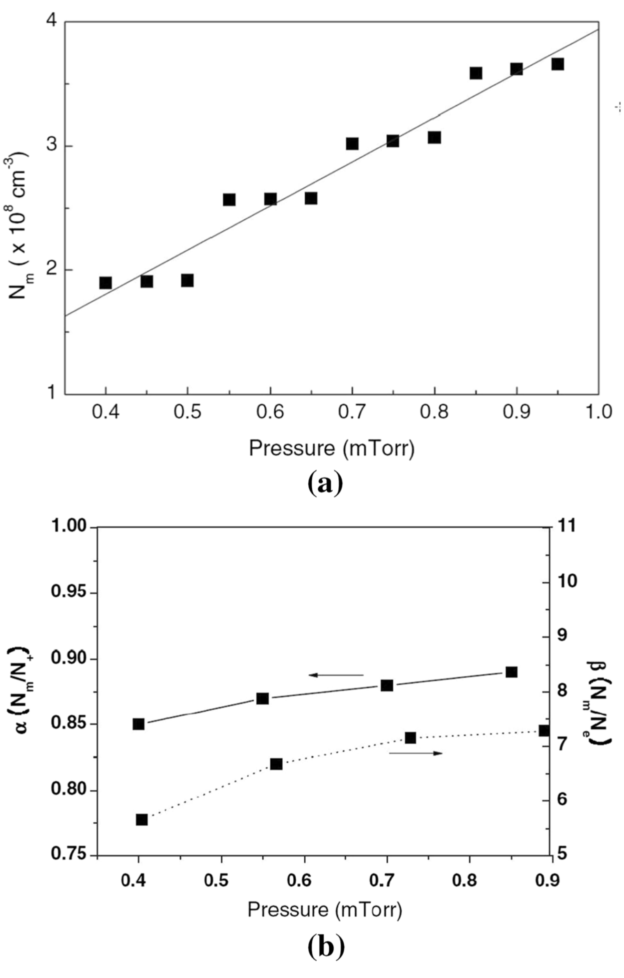


Fig. 11 Negative ion density and electronegativity deduced from probe measurements at an rf power of 300 W. Variations with pressure in the **a** negative ion density, and **b** electronegativity ($\alpha \equiv N_m/N_+$) and the ratio of negative ion to electron density ($\beta \equiv N_m/Ne$). Here, N_m, N_+, and N_- are the negative ion density, positive ion density, and electron density, respectively

3.2 Optical emission spectroscopy: collision-radiative model with He I lines for electron density and temperature [68]

For the measurement of the electron density by using He I line emission, the $D \rightarrow P$ transitions are better suited for two reasons. First, the excitation transfer cross sections for allowed transitions are much larger than those for non-allowed transitions. Second, the excitation transfer is inversely proportional to the energy difference between levels and strongly depends on the plasma density. For example, the energy difference between the 3 ^{1}P and the 3 ^{1}D levels is only 0.013 eV while the corresponding quantity between the 3 ^{3}P and the 3 ^{3}D levels is 0.066 eV. For the n = 4 level, △E(4 ^{1}P=4 ^{1}D) is only 0.006 eV while △E(4 ^{3}P=4 ^{3}D) is even smaller. Thus, these transitions will be more sensitive to the plasma density than to the electron temperature. For these reasons, we measured He I line ratios of 728.13 nm (3s ^{1}S → 2p ^{1}P) / 667.82 nm(3d ^{1}D → 2p ^{1}P) to determine the electron density. Line ratios using S → P transitions are better suited to measure electron temperature. The contributions

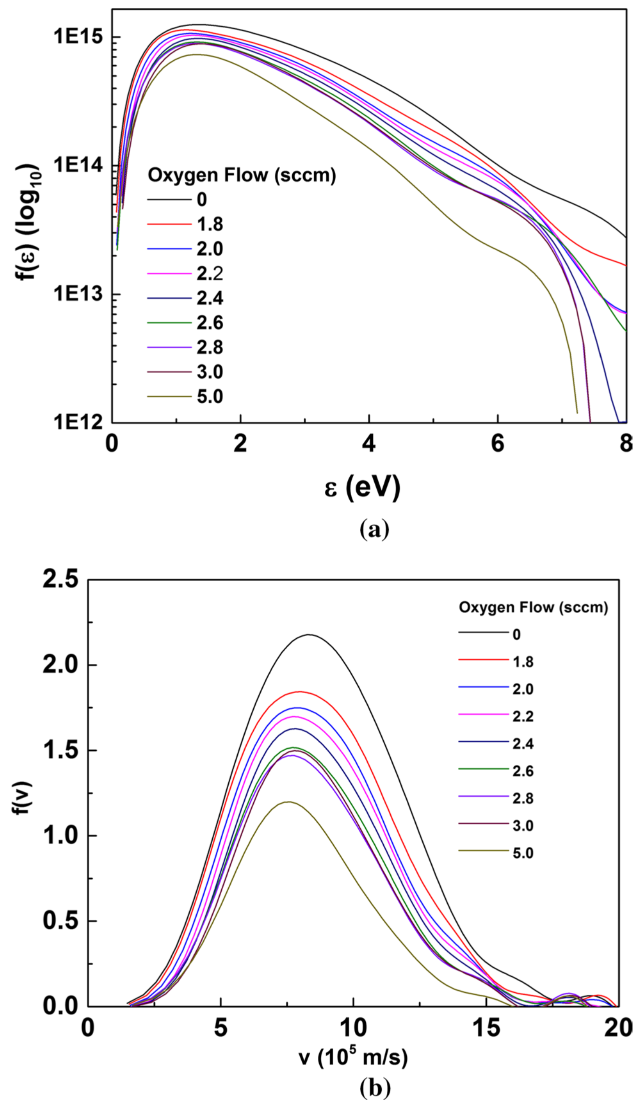


Fig. 12 **a** Electron energy distribution function $f(\varepsilon)$ on a logarithmic scale and **b** Gaussian fitting of the electron speed distribution function $f(v)$

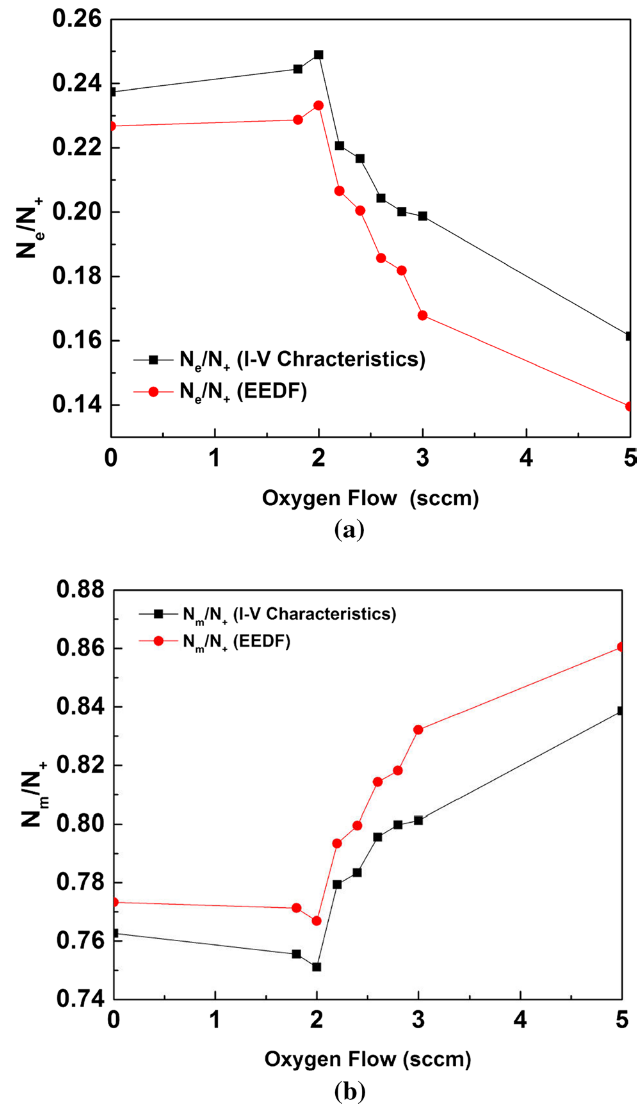


Fig. 13 Variations in the **a** ratio of the electron density (N_e) to the ion density (N_+), and the **b** ratio of negative ion density (N_m) to the ion density (N_+) with oxygen flow

from the meta-stable 2 ^{1}S and 2 ^{3}S states due to excitation transfer are small because these s− transitions are forbidden and the resulting cross section are small. Also, for a given n level, the energy of the S levels is significantly different from the energy of the other P, D, or F levels. Thus, the excitation transfer cross sections between S and any of the P, D, or F levels are also small compared to the cross section involving only P, D, and F levels. For these reasons, the 728.13 nm (3s ^{1}S → 2p ^{1}P) / 706.52 nm (3s ^{3}S → 2p ^{3}P) ratio is attractive for determining the electron temperature. Figure 14 shows the electron temperature and density profiles measured using the He I line intensity ratio and a single Langmuir probe. The electron temperature and density profiles estimated by using the former method, which is based on the collisional-radiative model, agree well with those measured by using a single Langmuir probe.

3.3 Laser-induced fluorescence (LIF) for Ar ion velocity distribution function [40, 69]

In the initial stage of LIF system development, one can use the Littrow external cavity laser, model of TEC-100, from Sacher Lasertechnik. The TEC-100 has the following specifications: maximum output power = 25 mW at 668.61 nm (at a −85 mA laser current), line width = 1 MHz, coarse tuning range = 665 – 675 nm with a rotating grating, fine tuning range = 0.34 nm with piezo-electric actuator control from 0 to 100 V, and a mode-hop free tuning range ≤ 16 GHz, with current coupling method. The schematic diagram

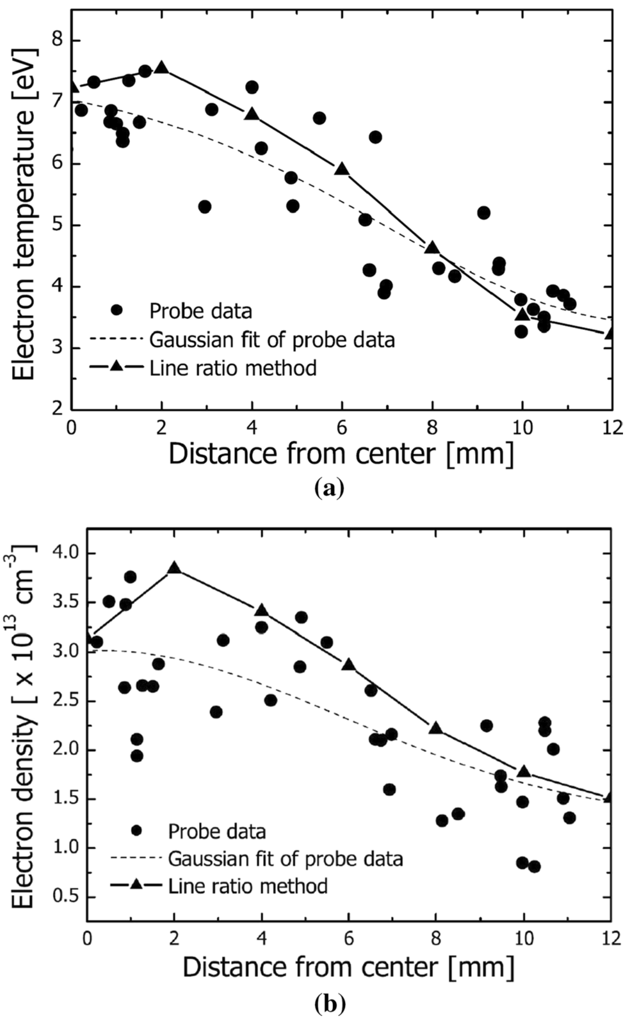
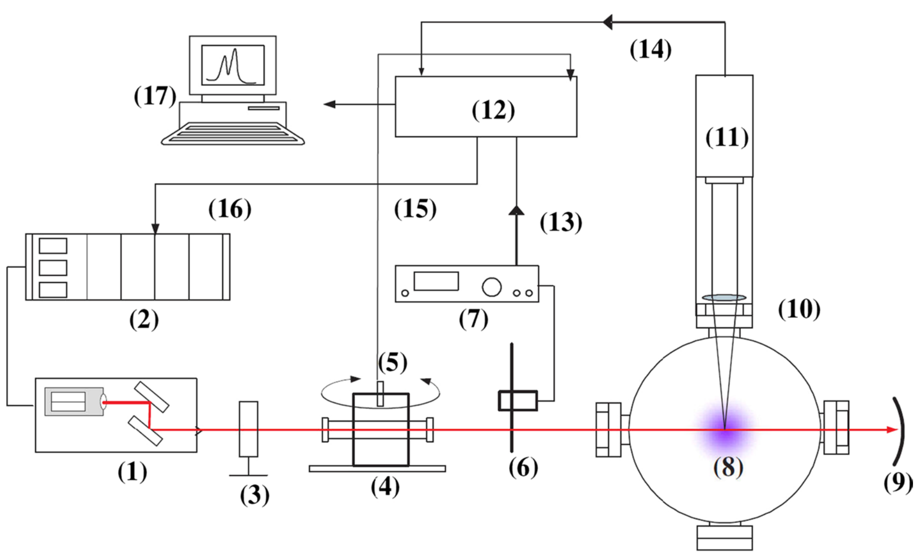


Fig. 14 Electron **a** temperature and **b** density profiles obtained using the He I line intensity ratio and a single Langmuir probe

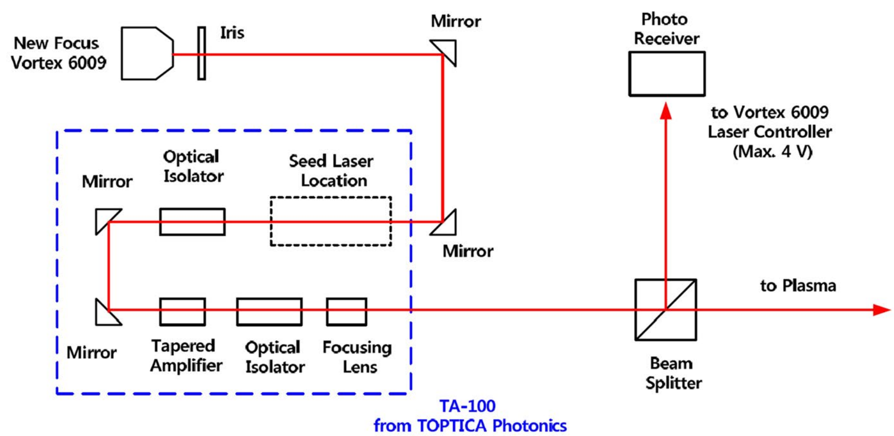
of this LIF system is shown in Fig. 15. To lower the laser power, one prevents the retro-reflection of the laser beam toward the diode laser by tilting the iodine cell less than 5 ° and using anti-reflection (AR) coated windows in all laser paths without an optical isolator. The iris is also located close to the diode laser to reduce the probability of retro-reflection of the laser beam back toward the laser diode. An iodine cell is generally used for *in-situ* measurements of the laser wavelength during LIF experiments. To obtain a clear iodine spectrum, one uses a commercially available silicon rubber heater to increase the iodine vapor pressure as was done by Keesee et al. [70]. The iodine cell, which is located in a thick black anodized aluminum housing, is heated to 90 ± 4°C, and the iodine cell spectrum is measured using NTE3032 photo-transistors. A 4 kHz mechanical chopper and a Stanford Research SR830 lock-in amplifier are used to collect the fluorescence signal by eliminating the electron impact induced radiation and the electronic noise. The SR830 lock-in amplifier is also used to measure the iodine cell spectrum with a NTE3032 photo-transistor through its auxiliary input and to ramp the input voltage of the piezo-electric actuator through its auxiliary output with a minimum resolution of 1 mV. The ramp voltage supplied to the laser is amplified by 10 times at the laser controller. The fluorescence signals are collected using the photomultiplier tubes (PMTs) and filtered using 1 nm bandwidth filter with a center wavelength of 442 nm after having been focused by using an imaging lens.

Later, the Littrow cavity laser was replaced by two lasers, a Littman/Metcalf laser and Master Oscillator Power Amplifier (MOPA) with a tapered amplifier for a wide mode-hop tuning range and higher laser power. The Littman/Metcalf laser, Vortex 6009 model, from New Focus (now combined to Newport Corporation) is used as a seed laser and MOPA, TA-100 model, from TOPTICA Photonics (former TUO

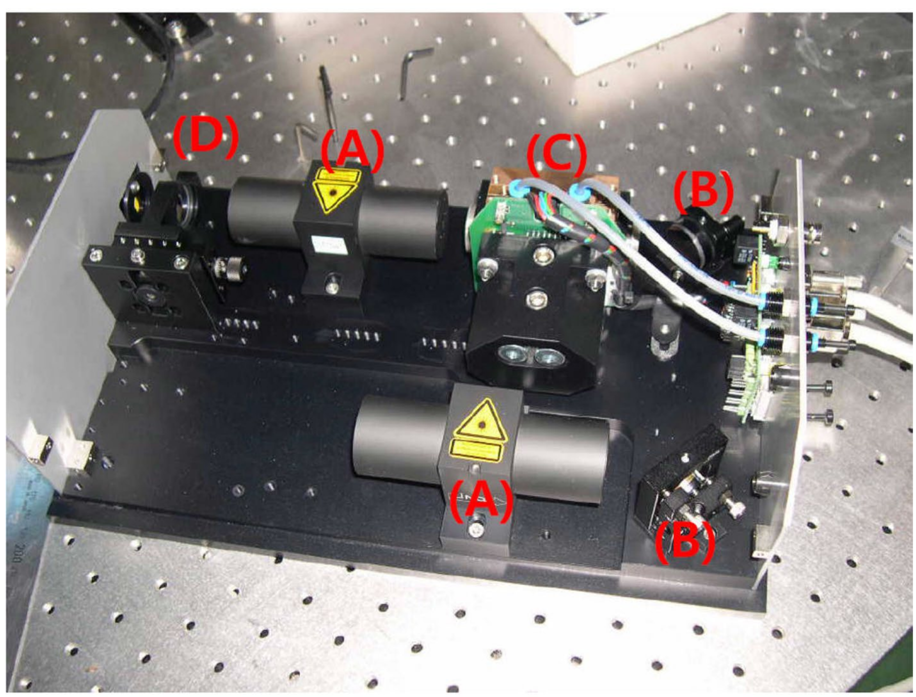
Fig. 15 Schematic diagram of the LIF system with an external Littrow cavity laser: (1) Littrow external cavity diode laser, (2) laser controller, (3) iris, (4) iodine reference cell, (5) NTE3032 photo-transistor, (6) chopper, (7) chopper controller, (8) plasma, (9) beam dump, (10) imaging lens, (11) PMT tube, (12) lock-in amplifier, (13) reference signal input, (14) fluorescence signal input, (15) iodine cell signal input, (16) voltage ramp, and (17) PC



Optics) is used for the power amplification. The Vortex 6009 model has the following specifications : maximum output power = 6 mW at 668.61 nm (at a -63 mA laser current), linewidth < 300 kHz, and a mode-hop free tuning range ≥ 70 GHz. The TA-100 model from TOPTICA Photonics is a full system for a tunable laser with a tapered amplifier: two stage optical isolator, alignment optics, and a seed laser with Littrow cavity configuration. However, one does not need the seed laser with the Littrow cavity configuration, which can be replaced by the Vortex 6009 Littman/Metcalf Laser. The power of Vortex 6009, seed laser, is changed during the wavelength scan. Therefore, one can use the laser current feedback control with a 2:1 beam splitter and photo receiver for constant power operation during a LIF scan because the LIF signal intensity is proportional to laser's power input. The integrated laser system with two diode lasers and a photograph of MOPA interior are shown in Fig. 16. Figure 16 shows photographs of the LIF system with Littrow cavity laser, and Littman/Metcalf cavity seed laser and MOPA configurations. Figure 17 shows the typical LIF line shape superimposed on Doppler and the Zeeman effects, which is measured at DiPS-1 with the magnetic field of 405 G and discharge current of 18 A.



(a)



(b)

Fig. 16 New laser system for the LIF: **a** schematic diagram of the tunable laser system for the LIF with a seed laser of Littman/Metcalf configuration and MOPA system, and **b** Photograph of MOPA Interior: **A** two stage optical isolator, **B** alignment mirror, **C** tapered amplifier, and **D** collimating lens

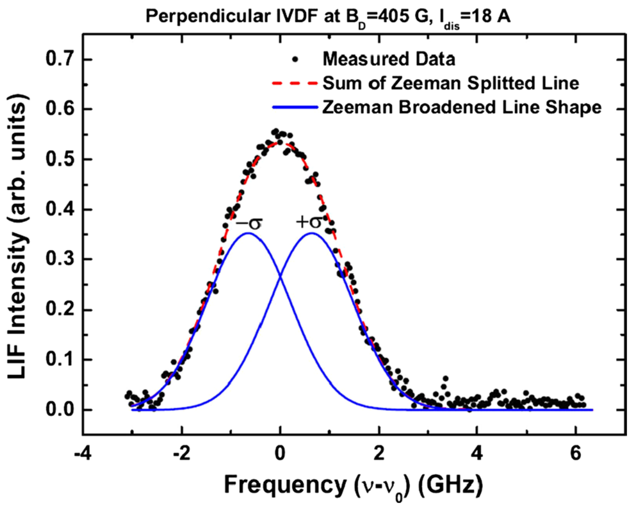


Fig. 17 LIF line shape due to superimposed Doppler and Zeeman effects

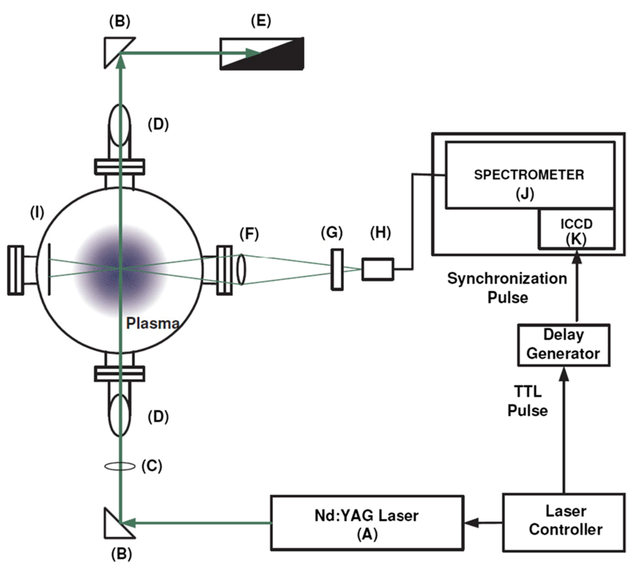


Fig. 18 Schematic diagram of the LTS system for DiPS-2 and scattering geometry: **A** frequency doubled Nd:YAG laser, **B** mirror array, **C** laser beam focusing lens (f = 700 mm), **D** brewster window and baffles, **E** beam dump, **F** collection lens, **G** polarizer, **H** optical fiber bundle, **I** viewing dump, **J** TGS, and **K** gated image intensified charge coupled device (ICCD)

3.4 Laser Thomson scattering (LTS) for the electron density and temperature

Figure 18 shows a schematic diagram of the LTS system for the DiPS-2. For maximum scattering probability, the polarization angle is fixed at $\varphi = 90°$. Also, the scattering angle is fixed at $\varphi = 90°$ to minimize stray light and to detect multiple points along the laser beam. For Thomson scattering experiments, one uses a frequency doubled (2ω), Q-switched Nd:YAG (Neodymium-doped Yttrium Aluminium Garnet: Nd:Y_3 Al_5 O_12) laser with a wavelength of $\lambda_i = 532$ nm, pulse width of $4 - 6$ ns, a pulse repetition rate of 20 Hz, and a pulse energy of 250 mJ (Continuum Surelite II-20). The laser beam is guided by using couples of dielectric mirrors and focused by using a plano-convex lens with a focal length of $f = 700$ mm. The laser's beam size is reduced by a focusing lens from 7 mm to less than 1 mm in the detection volume. For high transmission of the laser beam into the detection volume, one uses a Brewster window. Brewster window has a specific angle known as Brewster's angle, which is an angle incidence at light with a particular polarization is perfectly transmitted through a transparent dielectric surface with no reflection. The Brewster's angle is defined as $\theta_B = \arctan(\theta_1/\theta_2)$, where n_1 and n_2 are the refractive indices of typically air and dielectric medium, respectively. The Brewster's angle is $\theta_B = 57.12°$ for a quartz window, $n_2 = 1.547$ at 532 nm. In addition, the three stage baffles with hole diameter of 8 mm are located in the drift tubes for stray light rejection, and the beam dump is located at the end of laser path to prevent retro-reflection of the laser beam into detection volume. Stray light is regarded as optical noise produced by reflection from windows and the chamber wall. The light is collected by using an optical fiber bundle array through the 1:1 magnification image lens with $2f = 200$ mm. The polarizer is located in front of the optical fiber for the rejection stray light. The scattered photons are basically a polarized beam while stray light is un-polarized, so one can reduce stray light using a polarizer. The the optical fiber bundle array is composed of 71 optical fiber. Each optical fiber has a core diameter of 100 μm, a cladding thickness of 5 μm, and a buffer thickness of 7.75 μm with a numerical aperture of N/A=0.22 (or full acceptance angle of 25.4°). For plasma measurement, the optical fibers are separated by 63 fibers, 7 fiber × 9 position with displacement of 3.5 mm; in addition, 8 fibers are used for reference light inputs to check the spectrometer occasionally without breaking the optical alignments on the detection side. Also, the viewing dump is located at the opposite side of the detection optics for stray light reduction. The collected lights is dispersed by using a triple grating spectrometer and is presented by the image plane of the gated intensified charge coupled device (ICCD), which is composed of gated image intensifier and charge coupled device (CCD). The gated image intensifier, model M7971-81 model from Hamamastu Photonics, Inc., is composed of 2 stage multichannel plates (MCPs) and a GaAsP (Gallium, Arsenic, and Phosphorous) Photocathode, typically known as Generation III intensifier, with an aperture size of 18 mm for higher quantum efficiency, up to 50 % in the visible region. The CCD camera, model C4742-98ERG (ORCA-II) from Hamamastu Photonics, Inc., has 1344(H) × 1024(V) pixels, pixel size of 6.45 um(H) × 6.45 um(V), an effective area of 8.67

mm(H) × 6.60 mm(V), a cooling temperature of −60°C, and a dark current of 0.0045 electrons/pixel/s. A 2:1 relay lens is located between the MCPs and the CCD camera, so the real effective area is doubled. One can choice a typical gate width of 10 ns (min. 5 ns), which is fully covered by the laser's pulse width of 4-6 ns. If one considers a distance of 3 m from the laser head to detection volume, it can produced 10 ns of time delay which can not be ignored comparing with gate width. In addition, the coaxial cable and the electronics have some additional time delay. Therefore, the delay generator is used for synchronization between the laser pulse and the gate width of the image intensifier.

To reduce Rayleigh and stray light, one uses a triple grating spectrometer (TGS). [71] The TGS consists of three identical spectrometers, composed of 3 gratings, 6 lenses and 1 mirror. Although a diffraction grating with the groove densities of 1800 gr/mm with a size of 90 mm (H) × 90 mm (V) was used for TGS, one recommends a holographic grating rather than the a diffraction grating for higher signal to noise ratio. When the angle of the incident is $\theta_i = 15°$ at 1800 gr/mm, the diffraction angle is 44.32°, which is enough to arrange the optical elements without any interference. The spectral range is found to be 17.028 nm, which is desired for a DiPS-2 plasma and is related linear dispersion $\mathcal{L} = 0.013$ nm/pixel with $f = 400$ mm lens. For the lens, one uses an achromatic doublet rather than a plano-convex lens for reducing image aberration. One uses the 1 mm mask, which is made of black paper, for Rayleigh block. The entrance slit is fixed as 60 um for considering the misalignment of optical fiber bundle array. Although the misalignment is not significant, it does offered the image intensity slightly. The entrance slit width is narrower than the diameter of optical fiber bundle for rejection of stray light, which is generally formed in the edge region of optical fiber. The intermediate slit is fixed as 20 um for high rejection of stray light itself.

Figure 19 shows the results obtained using single probe and LTS measurements at a neutral pressure of 9.3 mTorr. The electron temperatures deduced from the probe measurement match well with those obtained from LTS measurements. For plasma density measurements, the plasma densities deduced by using the ion saturation currents agree well the results from the LTS measurements. On the other hand, the plasma densities deduced by using electron saturation currents are less than those obtained of LTS and ion saturation currents by factor of 2–3. Here, one does not consider the reduction factor because one does not know the sheath capacitance, so one considers only the collection areas $A_{pe} = 2 \times a_p l$ for magnetized electrons ($\rho_e \sim 0.06$ mm) and $A_{pi} = 2\pi a_p l$ for un-magnetized ions. Although one expects that the reduction in the electron density might be due to the reduction factors, more data, which can be obtained using different plasma densities (sheath capacitance) and difference magnetic fields (diffusion coefficients) for accurate reduction factor is to be found. Because electrons are magnetized deducing the electron density from the electron saturation current of the I–V characteristics would not be easy. If more data on magnetic field intensity could be acquired using probes and LTS, one would be able to the calibration factor (or reduction factor) from the electron saturation current; this remains as future work. Figure 20 shows the electron temperature and plasma density measured by using LTS and a single probe, especially the variation in the plasma density from ion saturation currents with pressure. Therefore, the ion-neutral collisions do not affect the single probe measurement, as one expected, because the pre-sheath is not collisionless. However, the data from the Mach probe is distorted by ion-neutral collisions so one can say that ion-neutral collisions do affect the Mach probe measurement when collision scale length $\lambda_{mfp}/L_{ps} \sim 1$ is considered, even though the plasma conditions, such as plasma density, electron temperature, neutral pressure, etc, are similar to the previous Mach probe measurement.

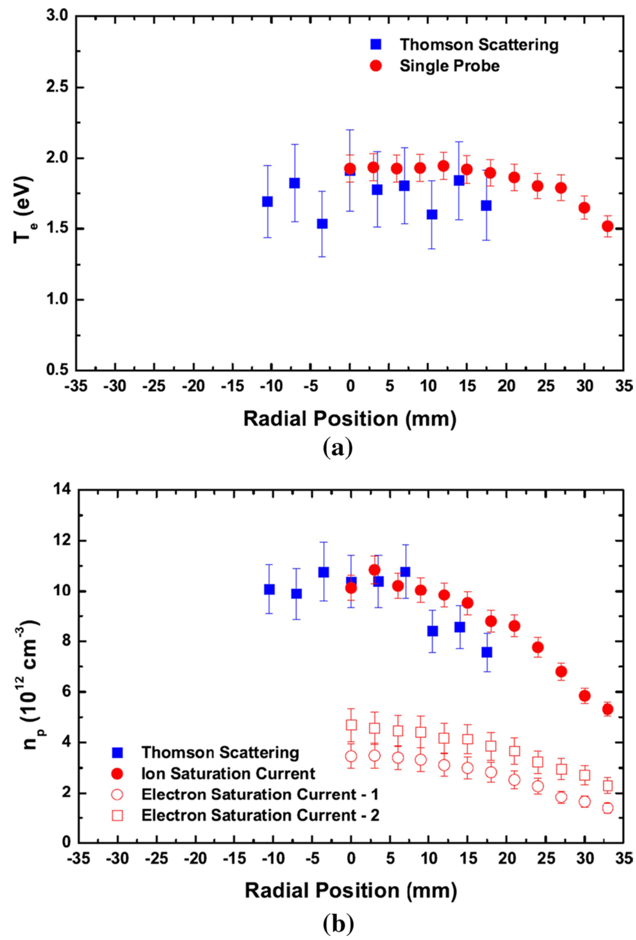
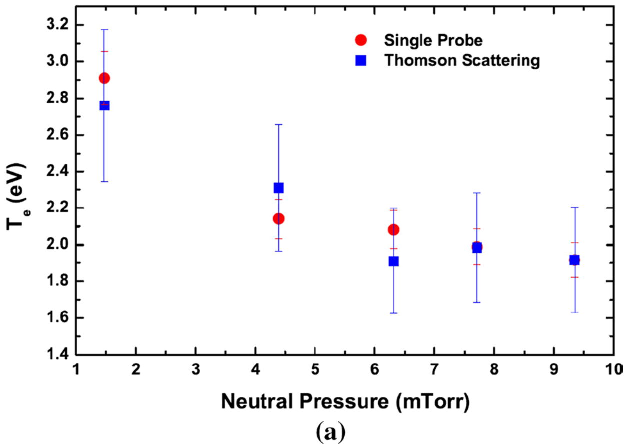


Fig. 19 **a** Electron temperature and **b** plasma density deduced from comparative measurements of LTS and single probe measured at a neutral pressure of 9.35 mTorr



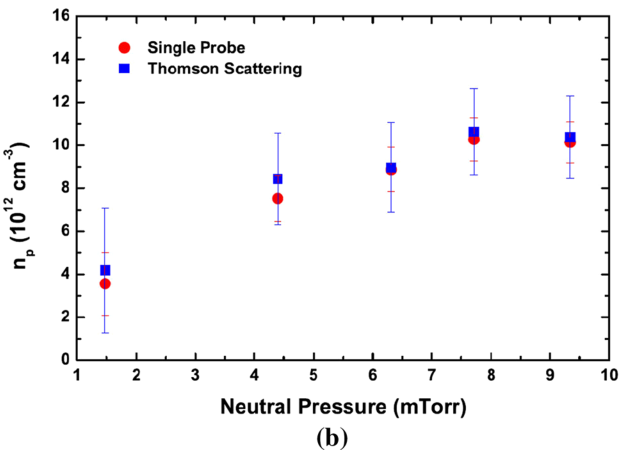


Fig. 20 Dependencies of the **a** electron temperature and **b** plasma density on the neutral pressure from LTS and single probe measurements. The plasma density from a single probe is used only to adopt the results for the ion saturation current

3.5 Laser photo-detachment (LPD) and laser Mie scattering (LMS) diagnostics for dust or negative ions

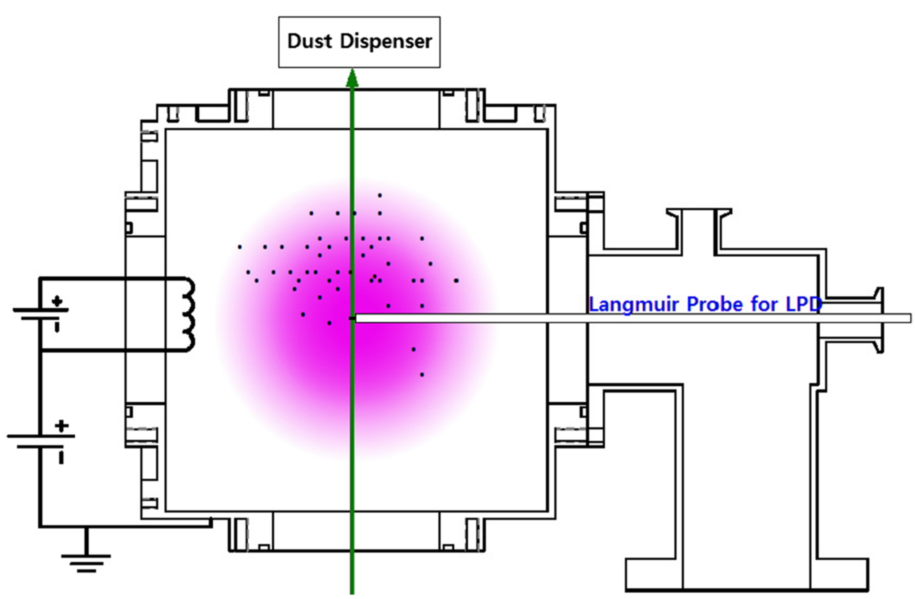
Laser-Photo Detachment (LPD) diagnostics are developed for negative ion and dust particle measurements. Figure 21 shows a schematic diagram of LPD diagnostic test device, which is composed of an Ar filament discharge plasma, a dust particle dispenser, a single Langmuir probe, and a Q-switched Nd:Yag Laser (max. 250 mJ at 532 nm). Figure 22 shows an example of LPD signal within dust particles. The dust particle density is given by

$$n_d = n_e \frac{\triangle I_e}{I_e} / Z_d = n_e \frac{\triangle I_e}{I_e} / C \frac{4\pi\epsilon_0 a T_e}{e^2} \ln \left[\frac{n_+}{n_e} \left(\frac{m_e T_i}{m_i T_e} \right)^{(1/2)} \right], \tag{29}$$

where n_e, I_e, n_d, Z_d, a, m_i, m_e, T_e, and T_i are the electron density, electron current, dust density, average number of charges carried by the dust particles, dust particle radius, ion mass, electron mass, electron temperature and ion temperature, respectively. In Eq. (29), C can be taken as a constant, and Matsoukas and Russell found that $C = 0.73$ gave the best fit to the numerical solution for an Ar plasma. [72]

For the measurement of dust particle size, the a laser scattering method, especially laser Mie scattering (LMS), is carried out for known dust particle sizes (or narrower size distribution): silver dust with radii of 0.25–0.5 μm and 0.5–1.5 μm. For Mie-scattering experiments, one uses 2nd harmonic pulsed Nd:YAG laser with an energy of 250 mJ (rep. rate of 20 Hz) and a fast photo-diode to collect scattered lights due to dust particles. The set-up of LMS experiments is similar to the LTS set-up, as shown in Fig. 18, except that the LMS uses a filtered photodiode while LTS uses a TGS system.

Fig. 21 Schematic diagram of the LPD



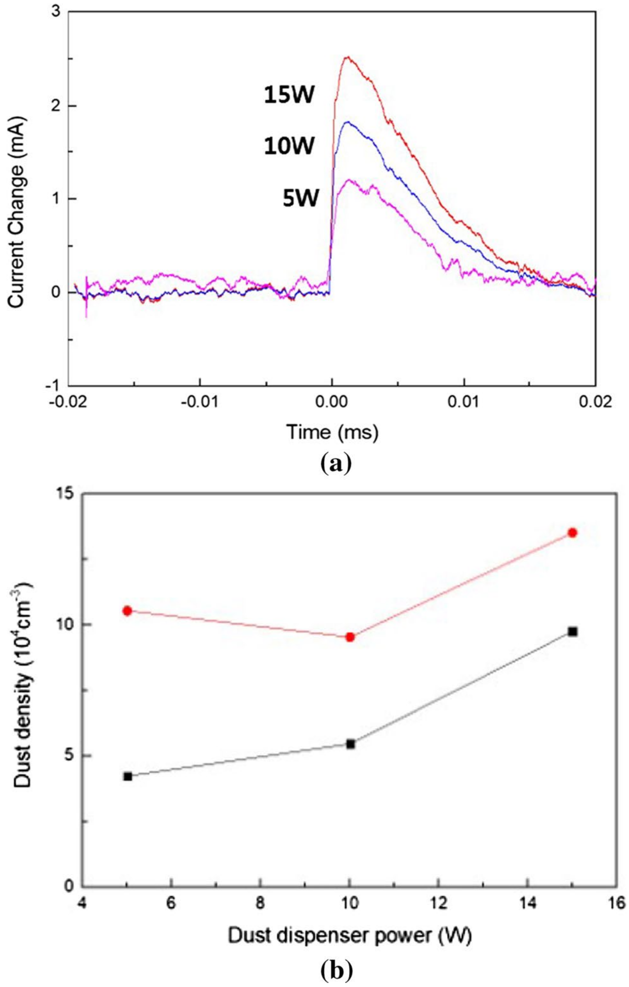


Fig. 22 LPD measurements of dust particles: Variation in the **a** signal with time for several dust dispenser powers and **b** the deduced dust density with dust dispenser power

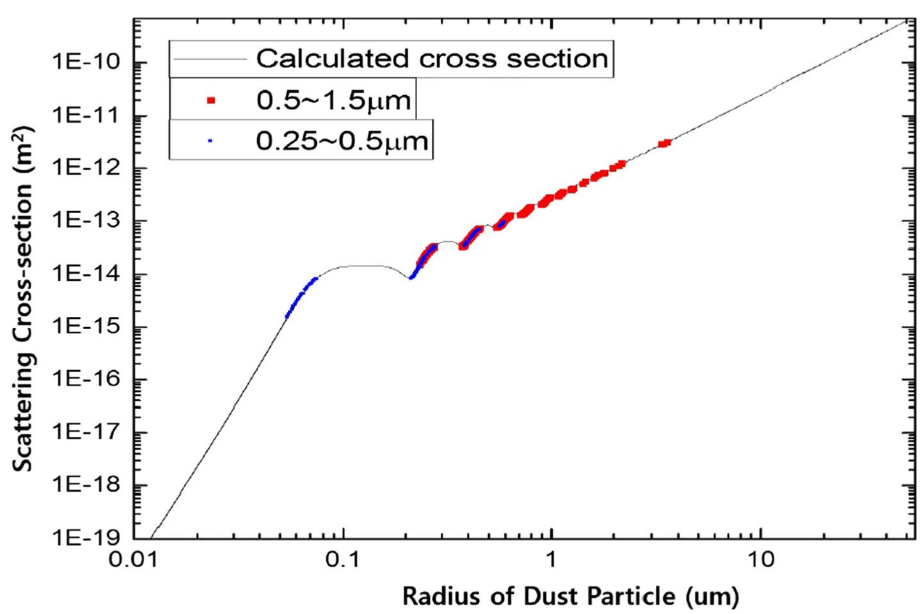
Figure 23 shows the Mie-scattering cross-section deduced from measurements on silver dust.

4 Summary

The Plasma-Material Interaction (PMI) in fusion devices is one of the key issues for the success of future fusion reactors. Generally, PMI should be analyzed by studying the plasma properties in the scape-off layer and divertor (limiter) regions, material erosion and deposition, impurity transport, fuel retention, and dust production and detection. Hanyang University has developed various plasma devices and plasma diagnostics through the **cEps** (Center for Edge Plasma Science) and the **cimpL** (Center for interaction of Materials and Plasmas) as part of the FCRC (Fuson Core Research Center) program and and the **DiPS** (Dust interaction of Plasma with Surfaces) as a research group, which are a part of the basic fusion research of Korean Fusion Program. Here, we summarize research in simulations of fusion edge plasmas in terms of linear plasma devices and their diagnostics.

The DiPS is composed of two plasma sources, LaB_6 DC cathode and helicon RF plasma, with the following objectives: divertor plasma simulation; space propulsion experimentation; processing/space/astrophysics simulations; and development of various plasma diagnostics. The DiPS-1 is part of the divertor plasma simulator of DiPS without space and processing simulators, where a low-power LIF is added and more electric probe diagnostics are augmented, is used only for the simulation of fusion edge and divertor plasmas. Although DiPS-1 can generate high density plasmas, which are high enough to test materials for the divertor region, covering the full spectrum of edge plasmas through SOL to

Fig. 23 Scattering cross-section for various radii of the dust particles as deduced by LMS experiment



divertor plate with impurity injection, the control of collisionality and flow is difficult because it has a shorter divertor simulation region, lower pumping capacity, no discrete section to control the neutral pressure, and small plasma size. The DiPS-1 has been upgraded to the DiPS-2, which has a magnetic nozzle for plasma flow control, dividing the multiple section to control the neutral pressure through highly differential pumping with a limiter structure, and upgrading the discharge power. DiPS-2 can generate plasma densities up to 10^{14} cm^{-3} ($T_e \sim 3$ eV) for Ar, 10^{13} cm^{-3} ($T_e \sim 5 - 10$ eV) for He & H_2 plasmas with plasma diameters of about 40 mm. For the dust transport, the TReD was also developed to validate the dust particle controls in fusion plasmas with a large area ICP source, a three-phase electrostatic curtain, and a dust collector.

To analyze the highly collisional and complex plasmas similar to those in the divertor region, a few electric probe theories have been developed, such as collisional Mach probe theory and electric probe theory for the negative ion density. Various plasma diagnostics also have been developed for electric probe calibration and for edge plasma properties: (1) OES with the He CR model is used to deduce the electron temperature, T_e, and density, n_e, in the fusion edge plasma. It is a non-invasive diagnostics (no-impurity source) with easy access and small space, and can give local plasma properties. (2) LIF is used to measure the Ar ion velocity distribution function (IVDF) and the Mach probe calibration for understanding edge plasma phenomena, such as near sonic plasma flows in the SOL. (3) LTS is used to measure the electron temperature, T_e, and the density, n_e, for the deduction of particle fluxes in multi-species plasmas like divertor region (impurity) which cannot be analyzed by using probes. (4) LPD and LMS diagnostics are used to measure negative ion and dust particles.

Although the DiPS-2 requires modification (or little upgrade) and more diagnostics, particularly for material diagnostics, such as laser-induced breakdown spectroscopy (LIBS) to analyze the retention of hydrogen isotopes, reflectometry to monitor the material damage versus exposure time, etc., it is a unique and very powerful linear plasma device in Korea, and could fully cover the PMI interactions through the main SOL to divertor plate, such as edge plasma transport, impurity transport, material physics, and particle and heat flux controls with impurity injection.

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